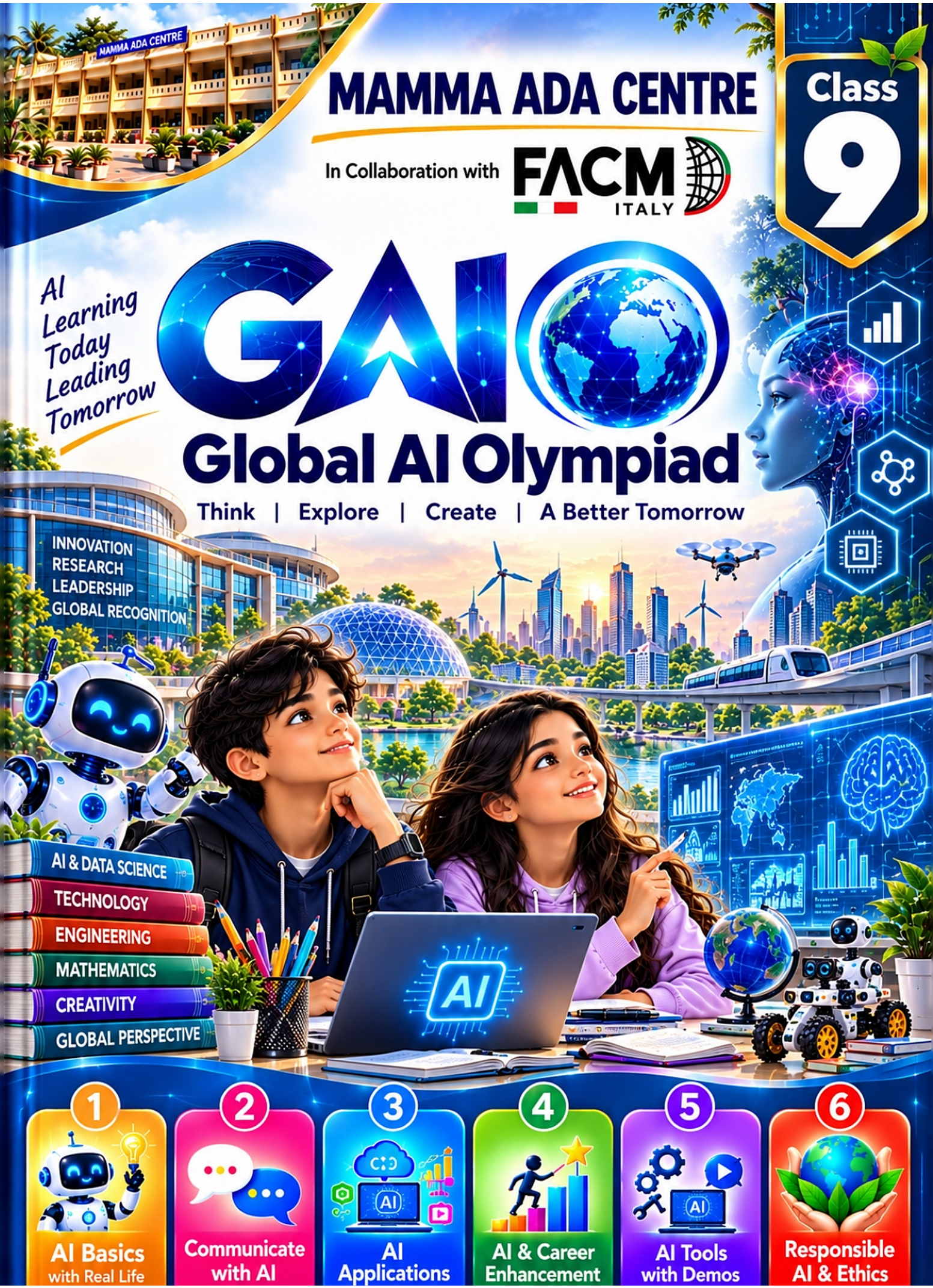

MAMMA ADA CENTRE
MAMMA ADA CENTRE
In Collaboration with FACM ITALY
Class 9
AI Learning Today Leading Tomorrow
GAIO
Global AI Olympiad
Think | Explore | Create | A Better Tomorrow
INNOVATION
RESEARCH
LEADERSHIP
GLOBAL RECOGNITION
AI & DATA SCIENCE
TECHNOLOGY
ENGINEERING
MATHEMATICS
CREATIVITY
GLOBAL PERSPECTIVE
AI
1 AI Basics with Real Life
2 Communicate with AI
3 AI Applications
4 AI & Career Enhancement
5 AI Tools with Demos
6 Responsible AI & Ethics

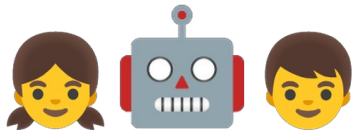
This book belongs to

Name: _____

Class & Section: _____

School: _____

📖 PREFACE



Artificial Intelligence is shaping how we learn, communicate, travel, farm, heal and create. As senior students, you are ready not only to use AI tools, but to understand how they work, question their answers and use them responsibly. This book is designed to build exactly those skills.

This book has been prepared for the Global AI Olympiad (GAIO) by Mamma Ada Centre, in collaboration with FACM, Italy. It follows the GAIO six-month journey: AI Discover, AI Connect, AI Solve, AI Rise, AI Create and AI Care. Each month introduces two topics suited to the curiosity and abilities of Class 9 learners.

Every topic opens with a clear introduction and continues with "Tech Talk", a group chat in which Riya, Kabir and their AI mentor Bolt explore ideas through real-life examples. A "How it works" flow, an India Spotlight, a hands-on Demo Lab with real AI tools, an "AI Chronicles" story, worksheets, quizzes, and standard and case-based exercises help you learn by reading, doing, questioning and creating.

Values are at the heart of this book. Along with learning what AI can do, children learn that people create and guide technology, that AI can make mistakes, and that safety, kindness and privacy matter online and offline.

🤝 A note for teachers and parents

Encourage students to try the Demo Labs under supervision, discuss the case studies, and question AI answers rather than accept them blindly. Praise reasoning and curiosity as much as right answers. An answer key is provided at the end of the book.

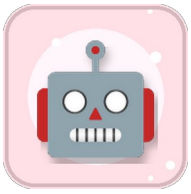
We hope this book sparks curiosity, creativity and confidence in every young learner. Curious minds, brighter futures!

With best wishes,

The GAIO Academic Team

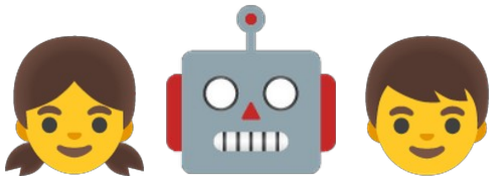
Mamma Ada Centre • In collaboration with FACM, Italy

Welcome, Seniors!



MEET BOLT!

Welcome, Class 9! I'm Bolt. This year: the DIKW ladder, prediction models, Python dictionaries, AI for the planet, research skills and critical thinking. Let's lead!



Look for these pictures in your book:

TECH TALK Concepts in a group chat

AI CHRONICLES Stories of teens & AI

LOOK AROUND YOU See AI in real life

LET'S DO IT! Do a fun activity

WORKSHEET Write, tick and draw

PUZZLE FUN Solve puzzles

DEMO LAB Hands-on AI tool demo

QUIZ TIME Answer quiz questions

TRUE OR FALSE True or false?

EXERCISE Standard & case-based Qs

MAGIC WORDS New words to learn

UNIT REVIEW Check your unit

My 6-Month AI Journey

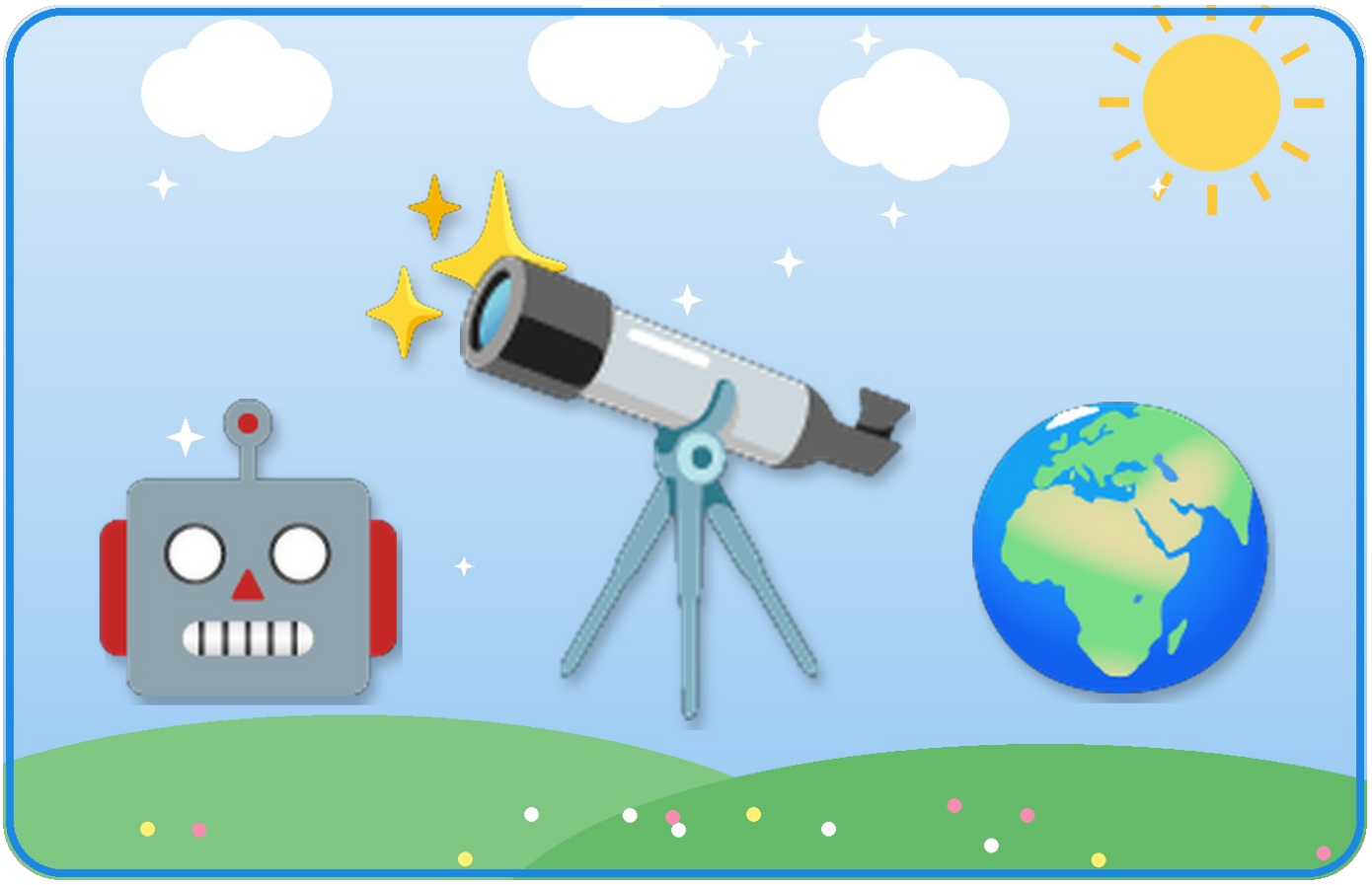
Follow the path. One month at a time!

MONTH 1 **AI DISCOVER** AI Basics	From Data to Intelligence • Machines That Predict
MONTH 2 **AI CONNECT** Communicate with AI	Python in Action • Solve It with Code
MONTH 3 **AI SOLVE** AI Applications	AI for Our Planet • AI for Public Good
MONTH 4 **AI RISE** AI & Career	Design Your AI Future • Skill-to-Career Map
MONTH 5 **AI CREATE** AI Tools	AI Research Room • AI Creator Studio
MONTH 6 **AI CARE** Responsible AI	Think Before You Believe • Digital Footprints & AI

MONTH 1

AI DISCOVER

See • Understand • Explore



Area: AI Basics

In this unit, you will explore:

Topic 1

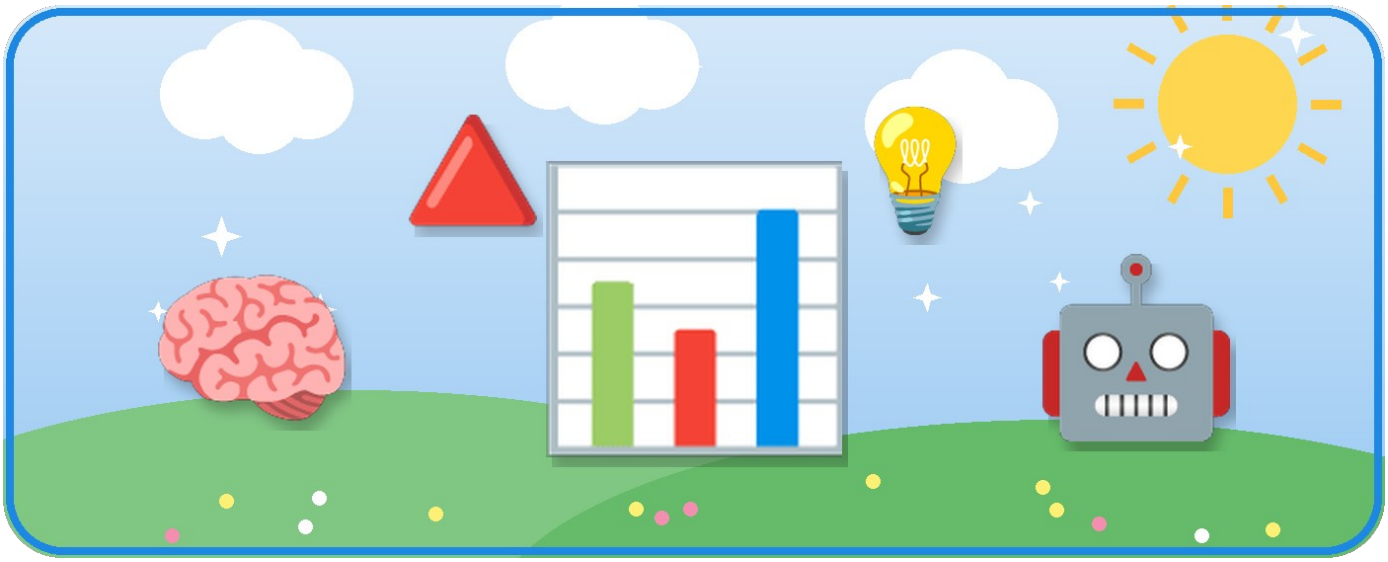
From Data to Intelligence

Topic 2

Machines That Predict

TOPIC 1 • MONTH 1 • AI BASICS

From Data to Intelligence



LET'S LEARN

- Data → Information → Knowledge → Wisdom.
- AI climbs this ladder by finding meaning in data.
- Raw numbers become smart decisions!

The DIKW pyramid

Data: raw facts

Information: organised

Knowledge: patterns

Wisdom: good decisions

MAGIC WORDS

Raw data
Unprocessed facts

Information
Organised data

Insight Useful discovery

💬 TECH TALK: From Data to Intelligence

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👦 **Kabir**
Bolt, our school canteen logs every sale. Is that intelligence 🤔?
10:02

🤖 **Bolt**
Not yet! That's raw data 🔢: '23 samosas, 11:05 a.m.'. Facts without meaning.
10:04 ✓✓

👧 **Riya**
When does it become useful?
10:06

🤖 **Bolt**
Organise it: 'Samosa sales peak at 11 a.m.' That's information 📊.
10:08 ✓✓

👦 **Kabir**
And knowledge?
10:10

🤖 **Bolt**
Understanding why: the long break starts at 11. Wisdom is acting on it: cook more by 10:50 💡.
10:12 ✓✓

👧 **Riya**

So AI climbs the DIKW ladder 🪜?

10:14

🤖 **Bolt**

Yes! It turns huge data into insights that help people decide 🎯.

10:16 ✓✓

👦 **Kabir**

Data information knowledge wisdom ✅!

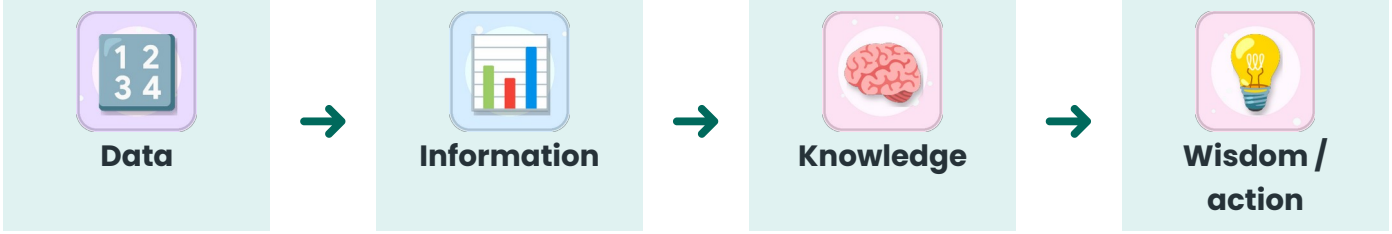
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Raw data becomes information when organised, knowledge when understood and wisdom when used to act. AI climbs this DIKW ladder to produce insights.

🔄 How it works

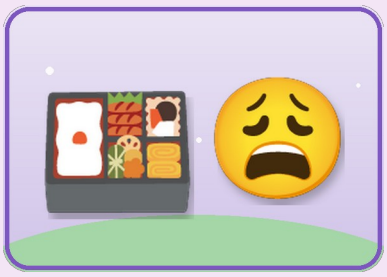


INDIA SPOTLIGHT

India's agricultural and weather data platforms turn raw sensor and satellite data into advisories that reach farmers by SMS.

AI CHRONICLES: The Canteen Code

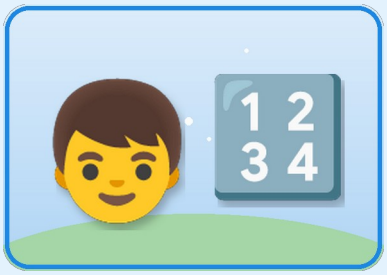
Kabir Riya Canteen Aunty



SCENE 1

Every day the canteen ran out of samosas by 11:10, while 40 sandwiches went to waste 🗑️.

Kabir collected the sales register: thousands of rows of raw data. It meant nothing yet.



SCENE 2

SCENE 3

Riya built a pivot table. Information appeared: samosas peaked at 11 a.m.; sandwiches sold only after sports.

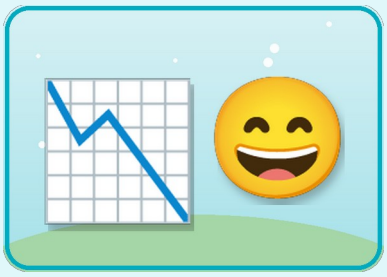
Knowledge: the long break began at 11, and sports teams ate after practice at 3 ⚽.

SCENE 4

SCENE 5

Wisdom: Aunty fried more samosas at 10:45 and made sandwiches at 2:30.

Waste fell by 70%, and sales rose. "Data talks if you listen," Aunty laughed 🎉.



SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
The team analyses library borrowing data	**Aunty throws away the register**	**The samosas start counting themselves**

🧐 Story check: answer the questions.

1. What was the raw data?

2. What knowledge did the team discover?

3. What action showed wisdom?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Data becoming intelligence

Temperatures → heatwave alert

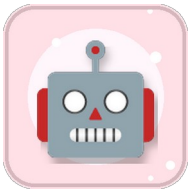
Sales → stock planning

Ball data → match strategy

Calls → ambulance placement

Marks → learning support

Rainfall → sowing advice



BOLT SAYS:

38, 39, 41... is just data. "A heatwave is coming, stay hydrated!" is intelligence.

FUN FACT!

The DIKW pyramid (Data, Information, Knowledge, Wisdom) is a famous idea in information science!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Build the DIKW Pyramid

You need:

Data cards

Pyramid chart

Teams

Steps:

1.
   Get cards with raw numbers: rainfall for 7 days.
2.
   Organise them into a table.
3.
   Find a pattern: is rain increasing?
4.
   Decide: what should a farmer do?

THINK & TALK

Where did the data become useful?

Draw what you did!

WORKSHEET: From Data to Intelligence (Part A)

Name: _____ Date: _____

1. Climb the DIKW ladder. Write 1, 2, 3, 4.

Wisdom ☐

Data ☐

Knowledge ☐

Information ☐

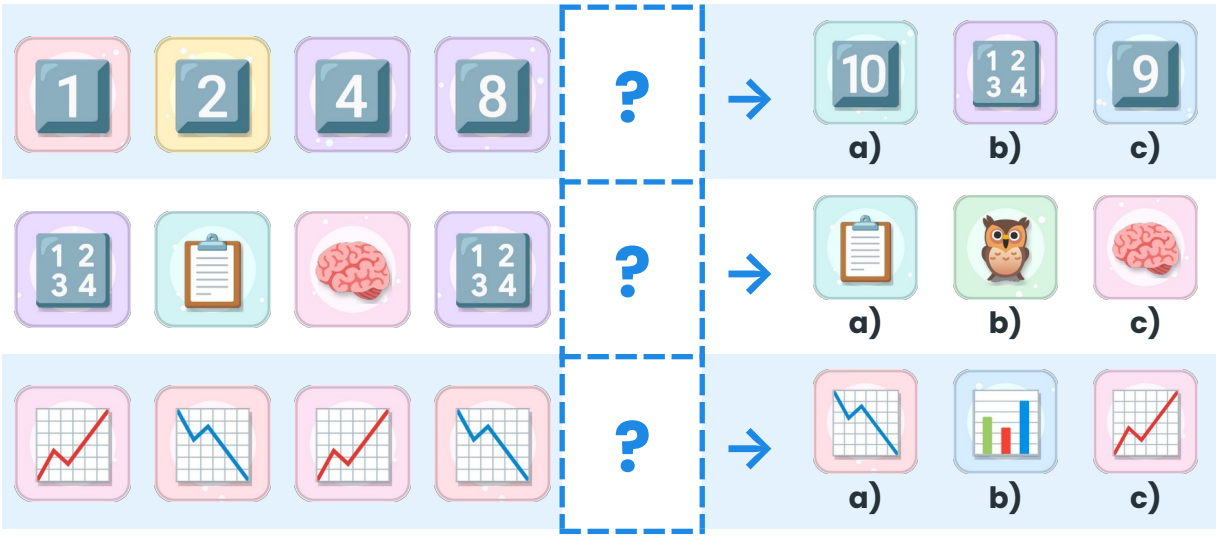
2. Match the example to the DIKW level.

32, 35, 38 ●	● Knowledge
Temp table by day ●	● Data
It's getting hotter ●	● Wisdom
Plan shade & water ●	● Information

WORKSHEET: From Data to Intelligence (Part B)

Name: _____ Date: _____

1. Data patterns! Circle what comes next.



2. Fill in the blanks. Use the word box.

WORD BOX: insight ✿ raw data ✿ information

Unprocessed facts are _____.

Organised data is _____.

A useful discovery is an _____.

3. Bonus: Draw your favourite picture from this topic!

I drew _____

PUZZLE FUN: From Data to Intelligence

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Raw data

Meaning: _____

Example: _____

Information

Meaning: _____

Example: _____

Insight

Meaning: _____

Example: _____

🧪 DEMO LAB: Spreadsheet (Google Sheets / Excel)

Name: _____ Date: _____

🎯 **Goal:** Turn raw canteen data into information, knowledge and a decision.

🔗 **Open:** sheets.google.com or Excel | 👩‍🏫 **With a teacher or parent**

👣 Steps

1	🔢	Teacher shares 2 weeks of raw canteen sales (item, time, quantity).
2	📊	Use a pivot table or SUMIF to total sales by item and by hour.
3	📈	Make a chart of sales per hour.
4	🧠	Explain WHY the peak happens (knowledge).
5	💡	Write one action the canteen should take (wisdom).

🔬 Record your observations

Best-selling item:	
Peak hour:	
Recommended action:	

💭 **Reflect:** Which step of the DIKW ladder needed human thinking the most?

QUIZ TIME: From Data to Intelligence

Circle the right answer!

Q1. Raw, unprocessed facts are...

a) Data

b) Wisdom

c) Toys

Q2. Organised data is...

a) Wisdom

b) Information

c) Waste

Q3. The top of the DIKW pyramid is...

a) Data

b) Information

c) Wisdom

Q4. A useful discovery is an...

a) Insight

b) Album

c) Icing

Q5. AI turns data into...

a) Balloons

b) Intelligence

c) Socks

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: From Data to Intelligence

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Data is raw facts.		
Wisdom is at the bottom of DIKW.		
Information is organised data.		
An insight is a useful discovery.		
Raw data is always useful as it is.		
AI turns data into intelligence.		

HOME CONNECT

Track your family's electricity meter for a week. Turn the data into one wise action!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: From Data to Intelligence

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Unorganised facts and numbers are ____.

(a) raw data ☐ **(b)** wisdom ☐ **(c)** insight ☐

2. Organised data with meaning is ____.

(a) noise ☐ **(b)** information ☐ **(c)** a virus ☐

3. Using knowledge to act well is ____.

(a) data ☐ **(b)** memory ☐ **(c)** wisdom ☐

4. DIKW stands for Data, Information, Knowledge and ____.

(a) Wisdom ☐ **(b)** Wi-Fi ☐ **(c)** Words ☐

B. Fill in the blanks. Use the word box.

insight organise why ladder

1. A useful discovery from data is an ____.

2. Pivot tables help ____ data.

3. Knowledge explains ____ something happens.

4. AI climbs the DIKW ____.

C. Write T for True and F for False.

1. Raw data alone always gives decisions. [ ]

2. Information is organised data. [ ]

3. Wisdom involves taking action. [ ]

4. AI cannot find insights. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Data	____	**(a)** Understanding why
2. Information	____	**(b)** Raw facts
3. Knowledge	____	**(c)** Acting wisely
4. Wisdom	____	**(d)** Organised facts

E. Answer in one word.

1. What is the first level of DIKW? _____

2. What is a meaningful discovery called? _____

3. What is the top level of DIKW? _____

F. Answer these questions.

1. Explain the DIKW ladder with a school example.

2. Why does turning data into wisdom need human judgement?

G. Case-based questions. Read and answer.

A canteen register showed raw sales. A pivot table revealed samosas peaked at 11 a.m. The team learned the long break starts then, so Aunty cooked earlier and waste fell 70%.

1. The pivot table produced ____.

(a) information ☐ **(b)** wisdom ☐ **(c)** raw data ☐

2. Knowing the break starts at 11 is ____.

(a) data ☐ **(b)** knowledge ☐ **(c)** noise ☐

3. Cooking earlier is ____.

(a) raw data ☐ **(b)** information ☐ **(c)** wisdom ☐

H. THINK DEEPER

Choose a dataset from your home (electricity bill, grocery list). Show how it can move up the DIKW ladder.

TOPIC 2 • MONTH 1 • AI BASICS

Machines That Predict

LET'S LEARN

- Prediction means estimating the future from past data.
- Regression predicts numbers. Classification predicts groups.
- No prediction is perfect. Errors must be measured!

Two prediction types

Regression: a number

Classification: a group

Error

Accuracy

MAGIC WORDS

Regression
Predicts a number

Classification
Predicts a group

Accuracy How often right

💬 TECH TALK: Machines That Predict

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya
Bolt, how does a food app predict delivery time 🛵?
10:02

🤖 Bolt
It learns from past orders: distance, traffic, time of day. That's prediction from data 📈.
10:04 ✓✓

👦 Kabir
Delivery time is a number. What kind of model is that?
10:06

🤖 Bolt
Regression: it predicts a number 🔢, like 27 minutes or tomorrow's temperature.
10:08 ✓✓

👧 Riya
And 'Will it rain: yes or no'?
10:10

🤖 Bolt
Classification: it predicts a group or label 🏷️, like rain/no rain or spam/safe.
10:12 ✓✓

👦 **Kabir**

How do we know if it's good?

10:14

🤖 **Bolt**

Measure errors 📏. Accuracy = correct ÷ total. For regression, check how far off predictions are.

10:16 ✓✓

👧 **Riya**

No prediction is perfect, so we measure and improve 🔁.

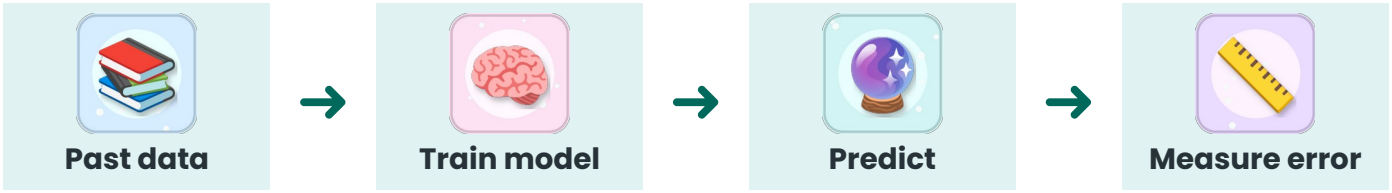
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Regression predicts numbers; classification predicts groups. Every prediction has error, so we measure accuracy and improve the model.

🔄 How it works

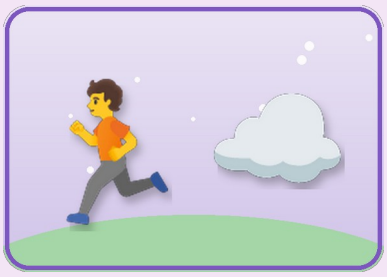


INDIA SPOTLIGHT

India Meteorological Department combines weather models and data to issue forecasts and early warnings for cyclones.

AI CHRONICLES: Will It Rain on Sports Day?

Riya | Sports Teacher | Bolt

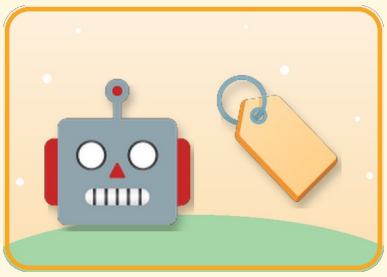


SCENE 1

Sports Day was in two weeks. Last year rain ruined it. Could AI help choose a date 📅?

Riya gathered 10 years of local weather data: humidity, pressure, wind and rain.

SCENE 2



SCENE 3

Bolt trained a classification model: Rain or No Rain. A regression model predicted rainfall in mm 🌧️.

Testing on last year's data: 82% accuracy. Good, but not perfect, so they planned a backup.

SCENE 4

SCENE 5

The model gave Saturday a 15% rain chance. They chose it and booked an indoor hall just in case.

Sports Day stayed sunny! "Predict, measure, and always have a plan B," said the teacher.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya predicts canteen demand next	**They ignore forecasts next year**	**The clouds join the race**

🧐 Story check: answer the questions.

1. Which model predicted rain/no rain?

2. What did the regression model predict?

3. Why did they book a backup hall?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Predictions around us

Rain chance: 70%

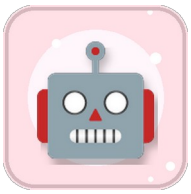
House price

Arrival time

Spam or not

Healthy or at risk

Will a customer buy?



BOLT SAYS:

I say "80% accurate". That means 20 out of 100 guesses may be wrong!

FUN FACT!

India's weather department uses computer models to forecast monsoon rains!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Height Predictor

You need:

Measuring tape

Graph paper

Classmates

Steps:

1.
   Measure 10 classmates: height and arm span.
2.
   Plot the points on a graph.
3.
   Draw a best-fit line through them.
4.
   Predict a new student's height from arm span!

THINK & TALK

How far was your prediction from the real height?

Draw what you did!

WORKSHEET: Machines That Predict (Part A)

Name: _____ Date: _____

1. Regression or classification? Tick ✓.

	Regression	Classification
Predict house price		
Spam or not spam		
Tomorrow's temperature		
Cat or dog photo		
Arrival time in minutes		

2. Fill in the blanks. Use the word box.

WORD BOX: accuracy ✿ regression ✿ classification

Predicting a number is _____.

Predicting a group is _____.

How often a model is right is its _____.

WORKSHEET: Machines That Predict (Part B)

Name: _____ Date: _____

1. Match the prediction to its type.

House price ₹50 lakh	●	●	Classification
Spam email	●	●	Regression
Arrival in 12 min	●	●	Classification
Healthy leaf	●	●	Regression

2. Building a predictor. Write 1, 2, 3, 4.

Measure accuracy ☐

Collect past data ☐

Train the model ☐

Predict new cases ☐

PUZZLE FUN: Machines That Predict

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Regression

Meaning: _____

Example: _____

Classification

Meaning: _____

Example: _____

Accuracy

Meaning: _____

Example: _____

🧪 DEMO LAB: Spreadsheet trendline / Google Colab (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Build a simple regression and classify results.

🔗 **Open:** Sheets chart trendline or colab.research.google.com | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Enter study hours (x) and test marks (y) for 15 students.
2		Insert a scatter chart and add a linear trendline.
3		Use the trendline to predict marks for 6 hours of study.
4		Classify each student: 'Pass' (≥ 40) or 'Needs help'.
5		Compare 3 predictions with actual marks and note the errors.

🔬 Record your observations

Predicted marks for 6 hours:	
Biggest prediction error:	
Is this regression or classification?	

💭 **Reflect:** Why can't study hours alone perfectly predict marks?

? QUIZ TIME: Machines That Predict

Circle the right answer!

Q1. Predicting a number is...

a) Regression

b) Classification

c) Painting

Q2. Spam or not spam is...

a) Regression

b) Classification

c) Music

Q3. 80% accurate means...

a) Never wrong

b) Always wrong

c) Wrong 20 in 100

Q4. Predictions use...

a) Past data

b) Only dice

c) Magic

Q5. A best-fit line helps...

a) Colour

b) Predict

c) Clean

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Machines That Predict

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Regression predicts a number.		
	Classification predicts a group.		
	Good predictions are never wrong.		
	Weather forecasts are predictions.		
	Predictions use magic, not data.		
	Accuracy shows how often a model is right.		

HOME CONNECT

Predict tomorrow's temperature with your family using a week of data. Check it tomorrow!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Machines That Predict

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Predicting a number like temperature is _____.

(a) regression ☐ **(b)** classification ☐ **(c)** clustering ☐

2. Predicting a label like spam/safe is _____.

(a) regression ☐ **(b)** classification ☐ **(c)** decoration ☐

3. Correct ÷ total gives _____.

(a) error ☐ **(b)** speed ☐ **(c)** accuracy ☐

4. Predictions are learned from _____ data.

(a) past ☐ **(b)** future ☐ **(c)** no ☐

B. Fill in the blanks. Use the word box.

perfect error regression classification

1. No prediction is _____.

2. The gap between predicted and actual is the _____.

3. A trendline is a simple _____ model.

4. Rain/No Rain is a _____ problem.

C. Write T for True and F for False.

1. Regression predicts numbers. [ ]

2. Classification predicts exact temperatures. [ ]

3. Errors should be measured. [ ]

4. A model with 82% accuracy is never wrong. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Regression	____	**(a)** Correct ÷ total
2. Classification	____	**(b)** Predicts numbers
3. Accuracy	____	**(c)** Prediction mistake
4. Error	____	**(d)** Predicts groups

E. Answer in one word.

1. Which model predicts numbers? _____

2. Which model predicts labels? _____

3. What measures correct predictions? _____

F. Answer these questions.

1. Give two examples each of regression and classification.

2. Why should we always measure prediction errors?

G. Case-based questions. Read and answer.

A model predicted Rain/No Rain with 82% accuracy, and another predicted rainfall in mm. The school picked a date with 15% rain chance and booked a backup hall.

1. Rain/No Rain prediction is _____.

(a) classification ☐ **(b)** regression ☐ **(c)** sorting ☐

2. Rainfall in mm is predicted by _____.

(a) classification ☐ **(b)** regression ☐ **(c)** clustering ☐

3. The backup hall shows that _____.

(a) models are perfect ☐ **(b)** rain is impossible ☐ **(c)** predictions can be wrong ☐

H. THINK DEEPER

A model predicts exam marks with 95% accuracy using only attendance. Is it fair to use it to decide which students get extra help? Explain.

UNIT REVIEW: AI DISCOVER

Name: _____ Date: _____

Circle the right answer!

Q1. Top of the DIKW pyramid:

a) Wisdom | b) Data | c) Information

Q2. Predicting a number is...

a) Classification | b) Regression | c) Art

Q3. Organised data is...

a) Wisdom | b) Waste | c) Information

Q4. Cat or dog photo is...

a) Classification | b) Regression | c) Music

Q5. How often a model is right:

a) Balloon | b) Accuracy | c) Cake

Q6. A useful discovery is an...

a) Album | b) Icing | c) Insight

WORD SEARCH: AI DISCOVER

🔍 **Find these words. They go across → or down ↓.**

RAW DATA

INSIGHT

REGRESSION

ACCURACY

DATA

PREDICT

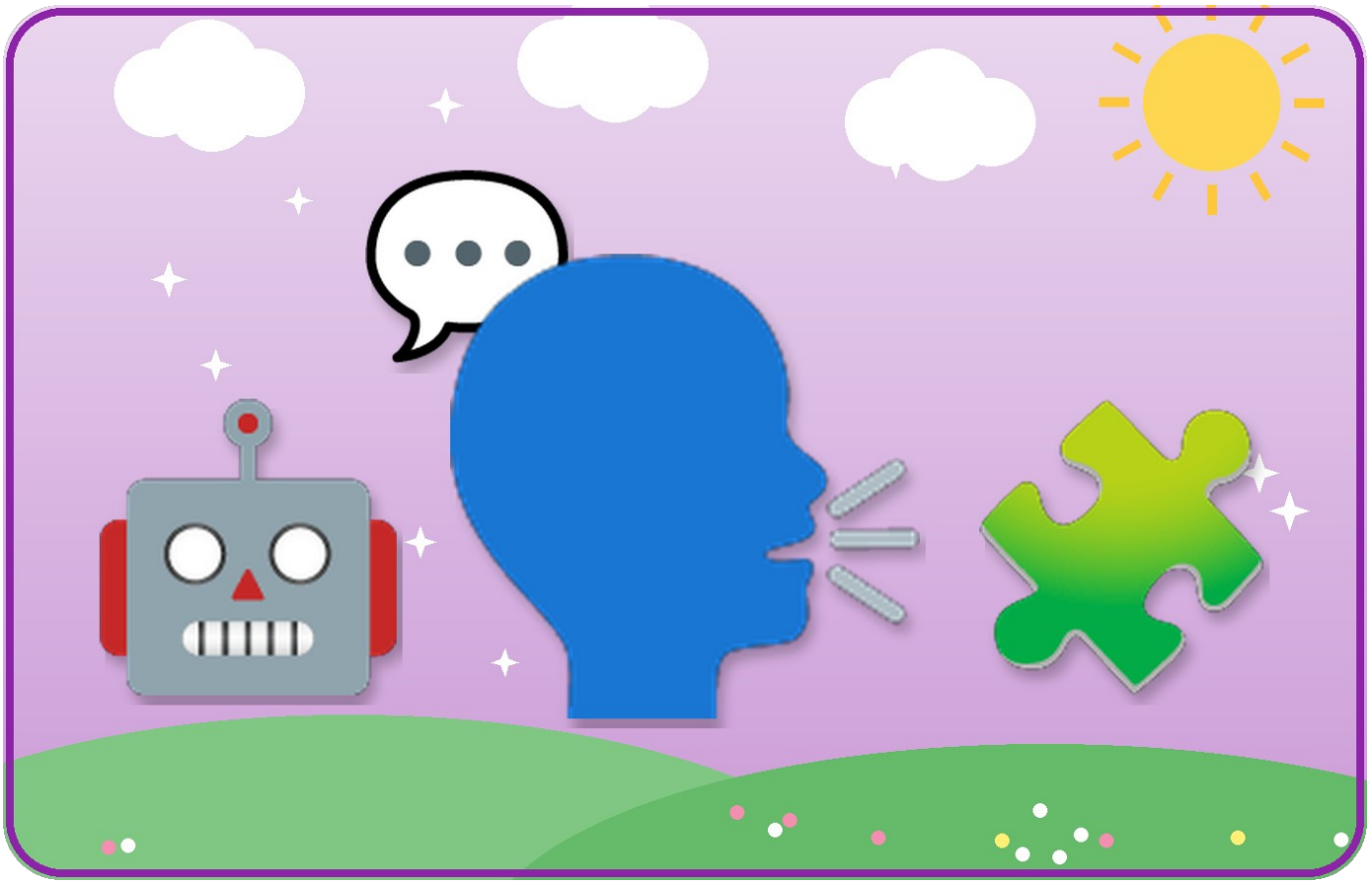
F	K	D	L	S	O	S	G	K	K
R	E	G	R	E	S	S	I	O	N
U	R	S	N	U	B	R	H	N	O
I	N	S	I	G	H	T	F	M	T
M	D	C	P	S	D	F	S	N	M
R	A	A	R	B	K	W	U	U	N
F	T	F	R	A	W	D	A	T	A
S	A	C	C	U	R	A	C	Y	H
P	R	E	D	I	C	T	A	G	T
T	H	N	S	M	U	M	P	I	T

I finished AI DISCOVER! Teacher's sign: _____

MONTH 2

AI CONNECT

Think • Instruct • Code



Area: Communicate with AI

In this unit, you will explore:

Topic 3

Python in Action

Topic 4

Solve It with Code

TOPIC 3 • MONTH 2 • COMMUNICATE WITH AI

Python in Action

LET'S LEARN

- Dictionaries store pairs: {"name": "Meera", "age": 14}
- Functions can take inputs: def area(l, b):
- while loops repeat until a condition is false.

Python power-ups

dict { }

def f(x)

while

return

MAGIC WORDS

Dictionary Key-value pairs

Parameter Input to a function

Return Send back a result

💬 TECH TALK: Python in Action

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 **Kabir**

Bolt, I need to store each student's name, age and house 🏠.

10:02

🤖 **Bolt**

Use a dictionary 📖: student = dict(name='Meera', age=14, house='Blue')

10:04 ✓✓

👧 **Riya**

How do we read one value?

10:06

🤖 **Bolt**

student['name'] gives 'Meera'. Keys unlock values 🔑.

10:08 ✓✓

👦 **Kabir**

Can functions take inputs?

10:10

🤖 **Bolt**

Yes, parameters! def area(l, b): return l * b. Then area(5, 3) returns 15 📐.

10:12 ✓✓

👧 **Riya**

What's the difference between print and return?

10:14

🤖 **Bolt**

print shows a value; return sends it back so you can use it later ↩️.

10:16 ✓✓

👦 **Kabir**

And a while loop repeats until a condition is false 🔁!

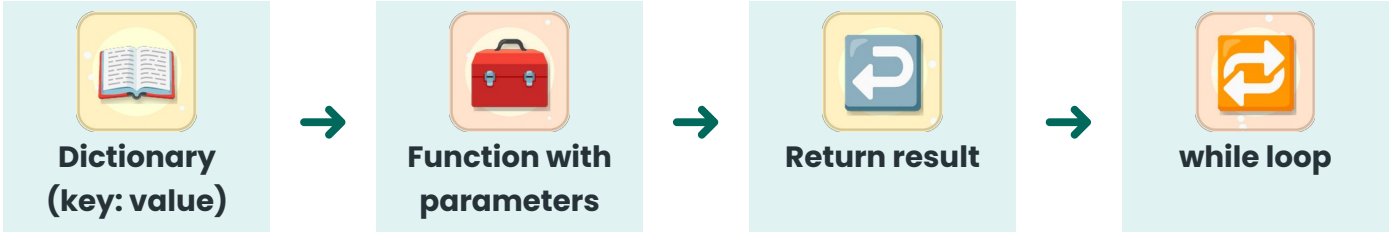
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Dictionaries store key-value pairs. Functions take parameters and return results. while loops repeat until a condition becomes false.

🔄 How it works



INDIA SPOTLIGHT

Python is a core language in India's school computer science curriculum and in many university AI courses.

AI CHRONICLES: House Points Tracker

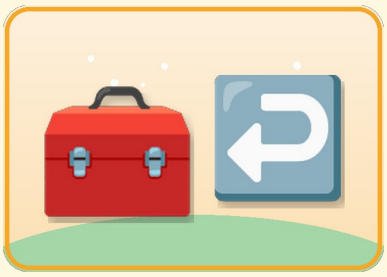
Kabir | Ms. Pillai | Python

SCENE 1

Ms. Pillai kept house points on paper. Totals were often wrong, and houses argued 😤.

Kabir made a dictionary: points = dict(Red=0, Blue=0, Green=0, Yellow=0).

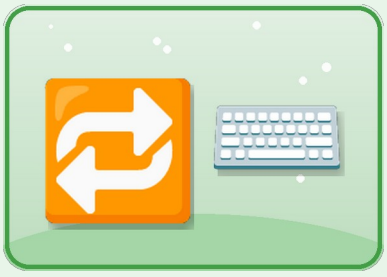
SCENE 2



SCENE 3

He wrote def add(house, n): that updated points and returned the new total.

A while loop kept asking for entries until someone typed 'done' ✅.



SCENE 4

SCENE 5

Bug: typing 'blue' in lowercase crashed it! He added .title() to fix the key.

Totals were now instant and fair. Ms. Pillai declared, "Python is our new scorekeeper!"

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir adds a chart of house points	**They go back to paper**	**The houses code their own points**

🧐 Story check: answer the questions.

1. What did Kabir store in a dictionary?

2. What did the add() function return?

3. What caused the bug, and how was it fixed?

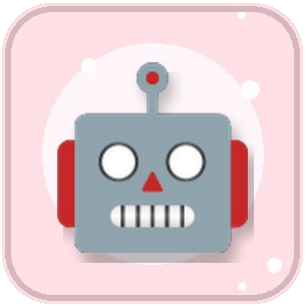
Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Python doing real work

Analysing survey data

Weather data charts

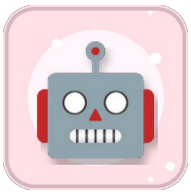


Training AI models

Building web apps

Game logic

Science simulations



BOLT SAYS:

return sends the answer back. def area(l, b): return l * b

FUN FACT!

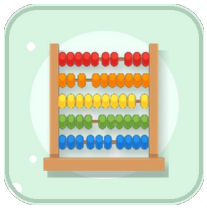
Python libraries like pandas and matplotlib help scientists analyse and plot data!

👀 **I spy AI! Draw one AI helper you saw today.**

LET'S DO IT! Paper Python Functions

You need:

Function cards



Calculator

Pairs

Steps:

1		Write: def double(x): return x * 2
2		Partner calls double(7). You return 14!
3		Make a dict: marks = {"Maths": 90, "Sci": 85}
4		Partner asks marks["Sci"]. Answer: 85!

THINK & TALK

Why are functions useful in big programs?

Draw what you did!

WORKSHEET: Python in Action (Part A)

Name: _____ Date: _____

1. What will Python output? Use the word box.

WORD BOX: 14 ✿ 90 ✿ 6

def double(x): return x*2 → double(7) → ____

marks = {"Maths": 90}; marks["Maths"] → ____

def area(l, b): return l*b → area(2, 3) → ____

2. Match the Python term to its meaning.

{ } ●	● Sends result back
def ●	● Repeats while true
while ●	● Dictionary
return ●	● Defines function

WORKSHEET: Python in Action (Part B)

Name: _____ Date: _____

1. Circle the correct Python code.

d = {"a": 1}

d = {"a" 1}

def f(x):

def f x:

while n > 0:

while n > 0

2. Complete the code. Use the word box.

WORD BOX: return ✿ def ✿ while

____ square(n):

____ n * n

____ count < 5: count += 1

PUZZLE FUN: Python in Action

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Dictionary

Meaning: _____

Example: _____

Parameter

Meaning: _____

Example: _____

Return

Meaning: _____

Example: _____

🧪 DEMO LAB: Online Python editor

Name: _____ Date: _____

🎯 **Goal:** Use dictionaries, functions with return values and while loops.

🔗 **Open:** any browser Python editor / Google Colab | 👩‍🏫 **With a teacher or parent**

👣 Steps

1	📖	student = dict(name='Meera', age=14, house='Blue'); print(student['house'])
2	✏️	student['age'] = 15; print(student)
3	🧰	def area(l, b): return l * b; print(area(5, 3))
4	🔁	n = 5; while n > 0: print(n); n = n - 1
5	🎯	Write def grade(m) that returns 'A', 'B' or 'C'.

🔬 Record your observations

Output of student['house']:	
Output of area(5, 3):	
Output of your while loop:	

💭 **Reflect:** Why is return more useful than print inside a function?

QUIZ TIME: Python in Action

Circle the right answer!

Q1. Key-value pairs are stored in a...

a) Dictionary

b) Tuple only

c) Balloon

Q2. An input to a function is a...

a) Gift

b) Parameter

c) Sock

Q3. return is used to...

a) Delete

b) Pause

c) Send back a result

Q4. while repeats until the condition is...

a) False

b) True

c) Musical

Q5. Python data library:

a) pizza

b) pandas

c) lions

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Python in Action

Look at the picture. Tick ✓ TRUE or FALSE.

	TRUE	FALSE
Dictionaries store key-value pairs.		
return sends a result back.		
while repeats until the condition is false.		
A parameter is an output of a function.		
pandas helps analyse data.		
Functions can never take inputs.		

HOME CONNECT

Make a Python-style dictionary of your family: name → favourite food!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Python in Action

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. A Python structure of key-value pairs is a ____.

(a) dictionary ☐ **(b)** list ☐ **(c)** tuple ☐

2. Inputs given to a function are ____.

(a) comments ☐ **(b)** parameters ☐ **(c)** errors ☐

3. To send a value back from a function, use ____.

(a) print ☐ **(b)** loop ☐ **(c)** return ☐

4. A ____ loop repeats until a condition is false.

(a) while ☐ **(b)** for-each ☐ **(c)** if ☐

B. Fill in the blanks. Use the word box.

key **parameters** **shows** **false**

1. student['name'] reads a value using its ____.

2. def area(l, b): has two ____.

3. print ____ a value; return gives it back.

4. A while loop needs a condition that becomes ____.

C. Write T for True and F for False.

1. Dictionaries store key-value pairs. [ ]

2. return and print do exactly the same thing. [ ]

3. A while loop can run forever if the condition never changes. [ ]

4. Functions cannot take parameters. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Dictionary	____	**(a)** Sends value back
2. Parameter	____	**(b)** Key-value pairs
3. return	____	**(c)** Repeats while true
4. while	____	**(d)** Function input

E. Answer in one word.

1. What is a function input called? _____

2. Which keyword sends a value back? _____

3. What unlocks a dictionary value? _____

F. Answer these questions.

1. Write a dictionary for a book with title, author and pages.

2. Explain the difference between print and return.

G. Case-based questions. Read and answer.

Kabir stored house points in a dictionary, wrote add(house, n) that returned the new total and used a while loop until 'done'. Typing 'blue' in lowercase crashed the program.

1. House names are the dictionary's ____.

(a) keys ☐ **(b)** loops ☐ **(c)** errors ☐

2. add() sends back the total using ____.

(a) print ☐ **(b)** return ☐ **(c)** while ☐

3. The crash happened because ____.

(a) Python was slow ☐ **(b)** the PC was old ☐ **(c)** the key's case did not match ☐

H. THINK DEEPER

Write a function that takes a dictionary of marks and returns the subject with the highest mark. Explain your logic.

TOPIC 4 • MONTH 2 • COMMUNICATE WITH AI

Solve It with Code

LET'S LEARN

- Computational thinking solves problems like a computer.
- Decompose, find patterns, abstract, design algorithms.
- Then code, test with many inputs and improve!

Computational thinking

Decompose

Patterns

Abstraction

Algorithm

MAGIC WORDS

Abstraction Keep only what matters

Test case Input to check code

Efficient Fast, less waste

💬 TECH TALK: Solve It with Code

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👧 **Riya**
Bolt, how do programmers solve big problems 🧩?
10:02

🤖 **Bolt**
With computational thinking: decompose, find patterns, abstract and design an algorithm 🧠.
10:04 ✓✓

👦 **Kabir**
What's abstraction?
10:06

🤖 **Bolt**
Keeping only important details. A metro map 🚇 hides streets but shows stations.
10:08 ✓✓

👧 **Riya**
Then we code?
10:10

🤖 **Bolt**
Yes, then test with many test cases 🧪: normal, edge and wrong inputs.
10:12 ✓✓

Kabir
Edge cases?
10:14

Bolt
Extreme values like 0 or a very large number ⚠️. Bugs love to hide there!
10:16 ✓✓

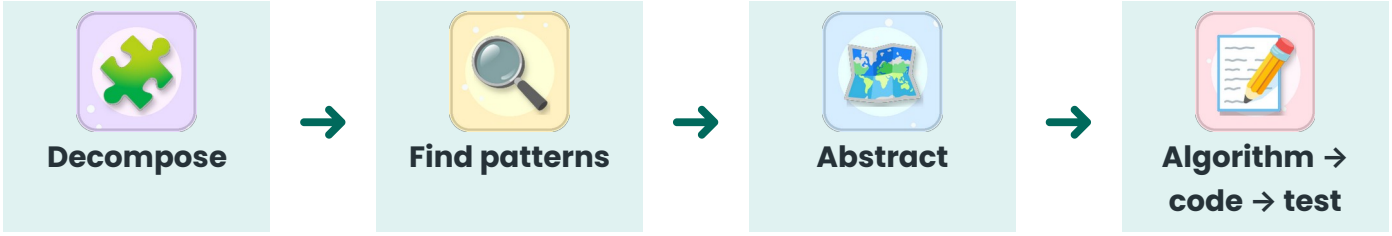
Riya
And efficient code uses fewer steps and less time ⚡.
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Computational thinking uses decomposition, pattern recognition, abstraction and algorithms. Test with normal, edge and invalid cases, and aim for efficient code.

🔄 How it works



INDIA SPOTLIGHT
India's school computing curricula and coding olympiads train students in computational thinking from middle school.

AI CHRONICLES: The Bus Route Puzzle

Riya

Kabir

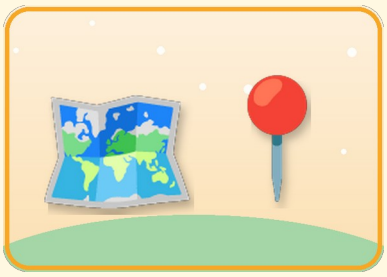
School Bus

SCENE 1

The school bus took 90 minutes to cover 12 stops. Students reached tired and late 😫.

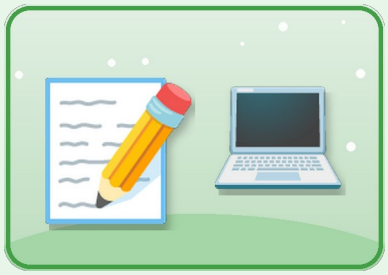
Riya decomposed the problem: stops, distances, timings and traffic.

SCENE 2


SCENE 3

Kabir abstracted the town into a simple map of dots and lines, ignoring shops and trees.

They designed an algorithm to always go to the nearest unvisited stop, then coded it.


SCENE 4

SCENE 5

Edge case: a road closed on Fridays! They added a rule to skip blocked roads 🚧.

The new route took 55 minutes. "Efficient code, happy students!" cheered the driver.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They optimise the evening route too	**The bus takes the long route again**	**The bus learns to fly**

🧐 Story check: answer the questions.

1. How did Kabir use abstraction?

2. What algorithm did they design?

3. What edge case did they find?

Rate this story: shade the stars.

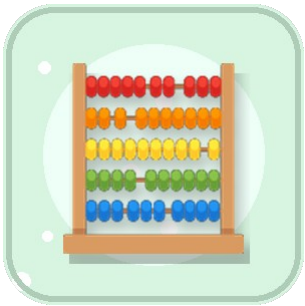
☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Problems students can code

Class election vote counter

Water bill calculator

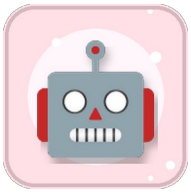
Library due-date tracker



GPA calculator

Password checker

Weather data analyser



BOLT SAYS:

Abstraction means ignoring details that don't matter. A metro map ignores the trees!

FUN FACT!

Computational thinking is useful in every subject, not just computers!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Vote Counter Program

You need:

Paper ballots

Code sheet

Class

Steps:

1		Class votes for a class leader on slips.
2		votes = {"Arun": 0, "Meera": 0}
3		for each slip: votes[name] += 1
4		Find the winner with max(). Check by hand!

THINK & TALK

How would your code handle a tie?

Draw what you did!

WORKSHEET: Solve It with Code (Part A)

Name: _____ Date: _____

1. Match the CT skill to the example.

Decompose ●

Pattern ●

Abstraction ●

Algorithm ●

● Steps to make tea

● Metro map without trees

● Split a project into tasks

● Every Monday is crowded

2. Good test case for a marks checker? Tick ✓.

	Useful	Not useful
marks = 50		
marks = 0		
marks = 100		
marks = pizza (no check needed)		

WORKSHEET: Solve It with Code (Part B)

Name: _____ Date: _____

1. Debug the vote counter. Use the word box.

WORD BOX: **+=** ✿ **{}** ✿ **max**

votes = ____ (empty dictionary)

votes[name] ____ 1

winner = ____(votes, key=votes.get)

2. Draw a flowchart for a water-bill calculator!

Start → … → End

PUZZLE FUN: Solve It with Code

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Abstraction

Meaning: _____

Example: _____

Test case

Meaning: _____

Example: _____

Efficient

Meaning: _____

Example: _____

DEMO LAB: Online Python editor

Name: _____ Date: _____

Goal: Design an algorithm, then test it with normal, edge and invalid cases.

Open: any browser Python editor / Google Colab | **With a teacher or parent**

Steps

1. Problem: find the largest number in a list without using max().
2. Write the algorithm in plain steps (pseudocode).
3. Code it with a for loop.
4. Test: [3, 9, 2]; [5]; [-4, -1, -9]; [] (empty list).
5. Fix any test that fails and note why.

Record your observations

Which test failed first?	
Why did it fail?	
Your fix:	

Reflect: Why do programmers test edge cases like empty lists?

? QUIZ TIME: Solve It with Code

Circle the right answer!

Q1. Keeping only what matters is...

a) Abstraction

b) Explosion

c) Rest

Q2. Splitting a problem is...

a) Painting

b) Decomposing

c) Singing

Q3. An input to check code is a...

a) Gift box

b) Suitcase

c) Test case

Q4. Fast with less waste is...

a) Efficient

b) Slow

c) Wasteful

Q5. CT is useful in...

a) Only computers

b) Every subject

c) Nothing

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Solve It with Code

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Abstraction keeps only what matters.		
	Decomposing means splitting a problem.		
	Testing with one input is always enough.		
	Efficient code wastes less.		
	A metro map is an example of abstraction.		
	Computational thinking is useless outside computers.		

HOME CONNECT

Use computational thinking to plan a family event: decompose, find patterns, make an algorithm.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Solve It with Code

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Keeping only important details is ____.

(a) abstraction ☐ **(b)** addition ☐ **(c)** attraction ☐

2. Breaking a problem into parts is ____.

(a) debugging ☐ **(b)** decomposition ☐ **(c)** deletion ☐

3. A step-by-step plan to solve a problem is an ____.

(a) error ☐ **(b)** emoji ☐ **(c)** algorithm ☐

4. Testing with extreme values checks ____ cases.

(a) edge ☐ **(b)** easy ☐ **(c)** empty-only ☐

B. Fill in the blanks. Use the word box.

abstraction **recognition** **steps** **cases**

1. A metro map is an example of ____.

2. Finding repeated features is pattern ____.

3. Efficient code uses fewer ____.

4. Inputs used to check code are test ____.

C. Write T for True and F for False.

1. Abstraction removes unnecessary details. [ ]

2. Edge cases never cause bugs. [ ]

3. Algorithms can be written before coding. [ ]

4. Testing only one input is enough. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Decomposition	____	**(a)** Keep key details
2. Pattern recognition	____	**(b)** Break into parts
3. Abstraction	____	**(c)** Step-by-step plan
4. Algorithm	____	**(d)** Find similarities

E. Answer in one word.

1. What is a step-by-step solution plan? _____

2. What hides unnecessary details? _____

3. What are extreme test inputs called? _____

F. Answer these questions.

1. List the four pillars of computational thinking.

2. Why should code be tested with edge cases?

G. Case-based questions. Read and answer.

Students turned their town into a dots-and-lines map, designed a 'nearest stop next' algorithm and found that a Friday road closure broke their route. The fix cut travel to 55 minutes.

1. The dots-and-lines map is ____.

(a) abstraction ☐ **(b)** decoration ☐ **(c)** decomposition ☐

2. 'Nearest stop next' is an ____.

(a) error ☐ **(b)** algorithm ☐ **(c)** edge case ☐

3. The Friday road closure was an ____.

(a) algorithm ☐ **(b)** abstraction ☐ **(c)** edge case ☐

H. THINK DEEPER

Design an algorithm to make your school timetable with no clashes. What would you abstract away?

UNIT REVIEW: AI CONNECT

Name: _____ Date: _____

Circle the right answer!

Q1. Key-value pairs are a...

a) Dictionary

b) Balloon

c) Pizza

Q2. Keeping only what matters:

a) Explosion

b) Abstraction

c) Rest

Q3. return is used to...

a) Delete

b) Pause

c) Send back

Q4. An input to check code is a...

a) Test case

b) Gift

c) Suitcase

Q5. def area(l, b): l and b are...

a) Balloons

b) Parameters

c) Cakes

Q6. Fast with less waste is...

a) Slow

b) Wasteful

c) Efficient

WORD SEARCH: AI CONNECT

🔍 **Find these words. They go across → or down ↓.**

DICTIONARY

PARAMETER

RETURN

TEST CASE

EFFICIENT

PYTHON

A	D	F	L	F	R	P	E	S	S
M	I	S	P	T	E	A	F	E	P
O	C	S	Y	T	T	R	M	F	U
M	T	M	T	E	U	A	P	F	F
N	I	P	H	S	R	M	S	I	H
R	O	I	O	T	N	E	R	C	S
S	N	M	N	C	P	T	P	I	M
U	A	T	P	A	R	E	H	E	L
F	R	W	I	S	R	R	K	N	K
S	Y	T	S	E	S	W	U	T	O

I finished AI CONNECT! Teacher's sign: _____

MONTH 3

AI SOLVE

Problems • Solutions • Impact

Area: AI Applications

In this unit, you will explore:

Topic 5

AI for Our Planet

Topic 6

AI for Public Good

TOPIC 5 • MONTH 3 • AI APPLICATIONS

AI for Our Planet

LET'S LEARN

- Climate change brings heatwaves, floods and droughts.
- AI helps predict disasters and track forests and oceans.
- It also helps us use clean energy wisely!

Planet-saving AI

Flood forecasts

Fire detection

Satellite maps

Solar & wind

MAGIC WORDS

Climate Long-term weather

Satellite Watches from space

Sustainable Good for the future

💬 TECH TALK: AI for Our Planet

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir

Bolt, floods hit Kerala, then heatwaves hit the north 🌡️. Is this climate change?

10:02

🤖 Bolt

Climate change makes extreme weather more frequent. AI helps us prepare 🌊.

10:04 ✓✓

👧 Riya

How?

10:06

🤖 Bolt

Satellites 🛰️ send images; AI spots forest loss, melting glaciers and flood risk.

10:08 ✓✓

👦 Kabir

Can it predict disasters?

10:10

🤖 Bolt

AI flood models can warn villages days ahead 📢, saving lives and cattle.

10:12 ✓✓

Riya
What about clean energy?
10:14

Bolt
AI forecasts sun and wind to balance solar ☀️ and wind 🌬️ power on the grid.
10:16 ✓✓

Kabir
But AI uses energy too, right 🤔?
10:18

Bolt
True! Sustainable AI means efficient models and green data centres 🌱.
10:20 ✓✓

😊 Type a message...

📌 **Pinned by Bolt**

AI uses satellite and sensor data to track climate change, warn of disasters and balance clean energy. AI itself must be sustainable.

🔄 How it works

INDIA SPOTLIGHT

ISRO's satellites provide data used to monitor floods, forests, crops and cyclones across India.

AI CHRONICLES: The Flood Warning

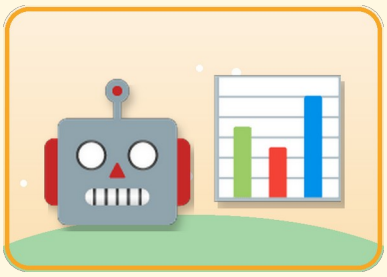
Kabir | Grandma | Satellite

SCENE 1

Kabir's grandmother lived near a river in Assam. Every monsoon, the family worried 😟.

High above, satellites captured river levels, rainfall and soil moisture every day.

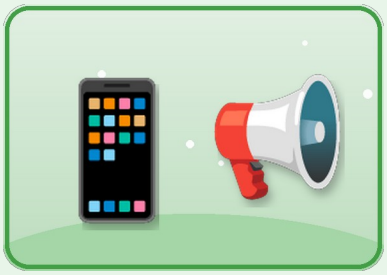
SCENE 2



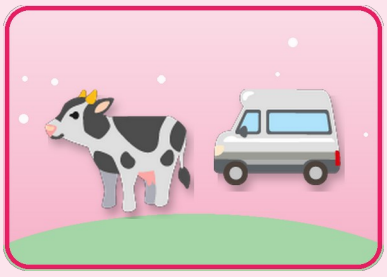
SCENE 3

An AI flood model combined the data and predicted the river would overflow in 48 hours ⚠️.

Alerts reached phones in the local language. Volunteers went door to door too.



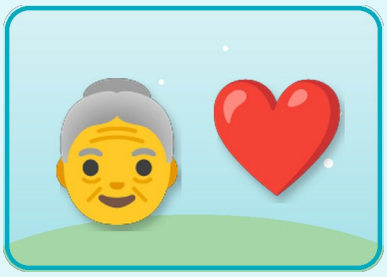
SCENE 4



SCENE 5

Grandma's family moved their cattle and valuables to higher ground in time ⛰️.

When the water came, everyone was safe.
"Machines watched the sky for us," Grandma said.



SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir builds a weather station at school	**The village ignores the next alert**	**The river apologises**

🧐 Story check: answer the questions.

1. What data did the satellites collect?

2. How did warnings reach people?

3. Why was the local language important?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: AI protecting Earth

Satellites track forests

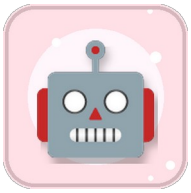
Flood warnings for villages

Tracking whales & fish

Balancing solar power

Drought-ready farming

Predicting air pollution



BOLT SAYS:

I can watch the whole planet from space. But saving it needs everyone!

FUN FACT!

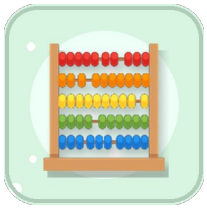
AI flood-forecasting systems now send warnings to people's phones in many Indian river regions!

I spy AI! Draw one AI helper you saw today.

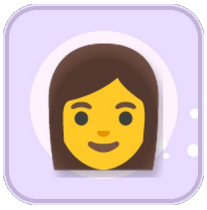
LET'S DO IT! Carbon Footprint Detective

You need:

Tracker sheet

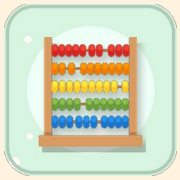


Calculator



Family

Steps:

1.
   Track one day: travel, electricity, food, waste.
2. 
   Score each: high, medium or low carbon.
3.
   Find your top 2 "high" habits.
4.
   Plan 2 changes for a greener week!

THINK & TALK

Which change would make the biggest difference?

Draw what you did!

WORKSHEET: AI for Our Planet (Part A)

Name: _____ Date: _____

1. Match the climate problem to the AI help.

Floods

Forest fires

Air pollution

Drought

Smoke spotted from space

Smart irrigation

Early phone warnings

Pollution forecasts

2. Fill in the blanks. Use the word box.

WORD BOX: satellite ✿ sustainable ✿ climate

Long-term weather is _____.

A _____ watches Earth from space.

Good for the future means _____.

WORKSHEET: AI for Our Planet (Part B)

Name: _____ Date: _____

1. Green habit or not? Tick ✓.

	Green	Not green
Cycling to school		
Lights on in empty rooms		
Recycling paper		
Long showers daily		
Planting trees		

2. Circle the clean energy sources.

Solar

Wind

Oil

Hydro

Coal

Geothermal

PUZZLE FUN: AI for Our Planet

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Climate

Meaning: _____

Example: _____

Satellite

Meaning: _____

Example: _____

Sustainable

Meaning: _____

Example: _____

🧪 DEMO LAB: NASA Worldview / Google Earth Engine timelapse (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Use satellite images to observe climate impacts.

🔗 **Open:** worldview.earthdata.nasa.gov | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Open Worldview and pick India.
2. Turn on the 'Fires and thermal anomalies' layer for a summer week.
3. Look at true-colour images of a major Indian river region before and after monsoon.
4. Compare a Himalayan glacier region over several years.
5. Write 3 observations and how AI could analyse them faster.

🔬 Record your observations

Fire hotspots seen:	
Flood/river change:	
Glacier change:	

💭 **Reflect:** How can AI-powered early warnings protect vulnerable communities?

? QUIZ TIME: AI for Our Planet

Circle the right answer!

Q1. Long-term weather is...

a) Climate

b) Time

c) Tune

Q2. AI satellites can spot...

a) Birthdays

b) Forest fires

c) Pizza

Q3. Good for the future is...

a) Wasteful

b) Polluting

c) Sustainable

Q4. Flood warnings can reach people's...

a) Phones

b) Balloons

c) Toys

Q5. Saving the planet needs...

a) Only AI

b) Everyone

c) Nobody

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI for Our Planet

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Satellites help track forests.		
	AI can help send flood warnings.		
	Oil is a clean, renewable source.		
	Sustainable means good for the future.		
	Only AI can save the planet.		
	Climate is long-term weather.		

HOME CONNECT

Do a carbon-footprint check for your family for one day. Choose 2 green changes!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI for Our Planet

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Long-term change in Earth's weather patterns is ____ change.

(a) climate ☐ **(b)** class ☐ **(c)** currency ☐

2. Images of Earth from space come from ____.

(a) drones only ☐ **(b)** satellites ☐ **(c)** submarines ☐

3. Meeting needs without harming the future is ____.

(a) wasteful ☐ **(b)** random ☐ **(c)** sustainable ☐

4. AI flood models give early ____.

(a) warnings ☐ **(b)** holidays ☐ **(c)** games ☐

B. Fill in the blanks. Use the word box.

solar **forests** **language** **energy**

1. AI balances ____ and wind power on grids.

2. ISRO satellites monitor floods and ____.

3. Alerts in the local ____ reach more people.

4. AI also uses ____, so it must be efficient.

C. Write T for True and F for False.

1. AI can analyse satellite images. [ ]

2. Climate change has no link to extreme weather. [ ]

3. AI itself uses energy. [ ]

4. Early warnings cannot save lives. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Satellite	____	**(a)** Good for the future
2. Early warning	____	**(b)** Images from space
3. Sustainable	____	**(c)** Balances power
4. Smart grid	____	**(d)** Alert before disaster

E. Answer in one word.

1. What gives images of Earth from space? _____

2. Which Indian agency launches satellites? _____

3. What alert comes before a disaster? _____

F. Answer these questions.

1. Give three ways AI helps fight climate change.

2. What does 'sustainable AI' mean?

G. Case-based questions. Read and answer.

An AI flood model used satellite rainfall and river data to predict an overflow 48 hours ahead. Alerts in the local language helped a village move people and cattle to safety.

1. The model used ____ data.

(a) satellite ☐ **(b)** shopping ☐ **(c)** music ☐

2. The warning came ____.

(a) after the flood ☐ **(b)** 48 hours before ☐ **(c)** a year later ☐

3. Local-language alerts made the system more ____.

(a) expensive ☐ **(b)** confusing ☐ **(c)** inclusive ☐

H. THINK DEEPER

Propose an AI project to reduce heatwave deaths in Indian cities. What data would it need and who would it help?

TOPIC 6 • MONTH 3 • AI APPLICATIONS

AI for Public Good

LET'S LEARN

- AI for public good helps people, not just profits.
- It supports education, health, farming and languages.
- It must be fair, safe and open for everyone!

AI for everyone

Language access

Farmer support

Learning for all

Accessibility

MAGIC WORDS

Public good
Benefits everyone

Inclusive Nobody left out

Accessible
Usable by all

💬 TECH TALK: AI for Public Good

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya

Bolt, does AI only help big companies make money 💰?

10:02

🤖 Bolt

No! AI for public good serves people: education 📚, health 🏥, farming 🌾 and languages.

10:04 ✓✓

👦 Kabir

Example?

10:06

🤖 Bolt

Translation AI lets a farmer ask questions in her own language and hear answers aloud 🗣️.

10:08 ✓✓

👧 Riya

What makes it 'for everyone'?

10:10

🤖 Bolt

Inclusive and accessible design ♿: works on basic phones, in many languages, for people with disabilities.

10:12 ✓✓

👦 **Kabir**

Who should check it's fair?

10:14

🤖 **Bolt**

Governments, experts and citizens. Public AI must be fair, safe and transparent ⚖️.

10:16 ✓✓

👧 **Riya**

AI that leaves no one behind 🤝!

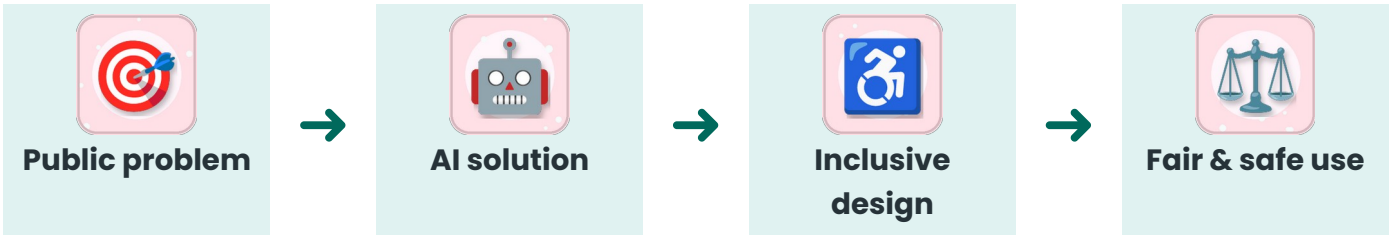
10:18

😊 Type a message...

📌 **Pinned by Bolt**

AI for public good helps people in education, health, farming and languages. It must be inclusive, accessible, fair and transparent.

🔄 How it works



INDIA SPOTLIGHT

Bhashini, India's national language AI mission, builds translation and speech tools for Indian languages.

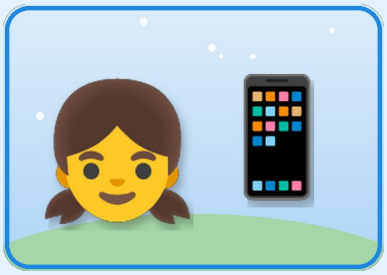
AI CHRONICLES: The Voice of the Village

Riya | Lakshmi Amma | Language AI

SCENE 1

Lakshmi Amma, a farmer, couldn't read English. Government crop advice online was useless to her 😕.

Riya showed her a language AI app that understood spoken Tamil 🗣️.



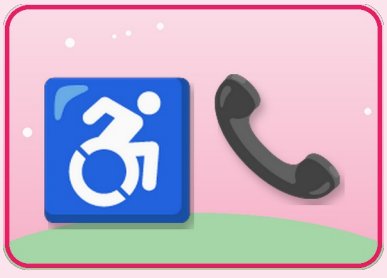
SCENE 2

SCENE 3

Amma asked about pests on her paddy. The app answered aloud in Tamil with simple steps.

One answer seemed wrong. Riya checked with the agriculture officer and reported the error.

SCENE 4



SCENE 5

They found it also worked on basic phones through a voice call. Accessible for all!

Soon 30 farmers used it. "AI should speak our language," Amma said proudly.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya helps add more dialects	**The app stays English-only**	**The paddy starts talking**

🧐 Story check: answer the questions.

1. Why was the online advice useless to Amma?

2. What did Riya do when an answer seemed wrong?

3. What made the app accessible?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Public-good AI in India

Translation across Indian languages

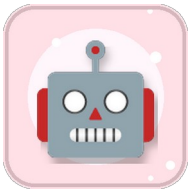
Crop advice by voice

Lessons in mother tongue

Rural telemedicine

Apps that read for the blind

Disaster relief planning



BOLT SAYS:
The best AI helps the person who needs help the most!

FUN FACT!
India's Bhashini mission builds AI tools to translate between Indian languages!

👀 **I spy AI! Draw one AI helper you saw today.**

LET'S DO IT! AI for Good Pitch

You need:

Pitch sheet

Poster paper

Teams

Steps:

1		Find a problem in your community.
2		Design an AI idea to help.
3		Check: is it fair, safe and inclusive?
4		Pitch it to the class in 2 minutes!

THINK & TALK

Who might be left out by your idea? How can you include them?

Draw what you did!

WORKSHEET: AI for Public Good (Part A)

Name: _____ Date: _____

1. Circle the AI-for-good examples.

Language translation

Gambling app

Reading app for the blind

Crop advice

Fake videos

Disaster planning

2. Fill in the blanks. Use the word box.

WORD BOX: inclusive ✿ accessible ✿ public good

Benefiting everyone is _____.

Nobody left out means _____.

Usable by all is _____.

WORKSHEET: AI for Public Good (Part B)

Name: _____ Date: _____

1. Match the community need to the AI help.

	Blind student		Translation
	Farmer		Reads text aloud
	Different language		Relief planning
	Cyclone		Crop advice

2. Design an AI app that helps your community!

My app helps: _____

PUZZLE FUN: AI for Public Good

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Public good

Meaning: _____

Example: _____

Inclusive

Meaning: _____

Example: _____

Accessible

Meaning: _____

Example: _____

🧪 DEMO LAB: Bhashini / a translation app (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Test how language AI includes (or excludes) people.

🔗 **Open:** bhashini.gov.in or any translation app | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Speak one sentence in your mother tongue and translate it to English.
2		Translate it back. Did the meaning change?
3		Try 3 more Indian languages.
4		Try the app's audio or text-to-speech features.
5		Rate accuracy for each language (1–5).

🔬 Record your observations

Best-performing language:	
Weakest language:	
Accessibility feature found:	

💭 **Reflect:** Why do some languages work better than others in AI tools? How can we fix this?

? QUIZ TIME: AI for Public Good

Circle the right answer!

Q1. AI for public good helps...

a) People

b) Only profits

c) Nobody

Q2. Nobody left out is...

a) Exclusive

b) Inclusive

c) Locked

Q3. Bhashini helps with...

a) Games

b) Cooking

c) Indian languages

Q4. Apps that read text aloud help...

a) Blind people

b) Fish

c) The Moon

Q5. Good public AI must be...

a) Unfair

b) Fair & safe

c) Secret

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI for Public Good

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Bhashini helps translate Indian languages.		
	Public-good AI is only for profit.		
	AI apps can read text for blind people.		
	Inclusive means nobody is left out.		
	Public AI should be fair and safe.		
	Public AI should be secret and unfair.		

HOME CONNECT

Ask your family: what local problem could AI help solve in our area?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI for Public Good

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. AI that helps people rather than only profits serves the public ____.

(a) good ☐ **(b)** bill ☐ **(c)** noise ☐

2. Design that works for people with disabilities is ____.

(a) exclusive ☐ **(b)** accessible ☐ **(c)** expensive ☐

3. India's national language AI mission is ____.

(a) UPI ☐ **(b)** Aadhaar ☐ **(c)** Bhashini ☐

4. Public AI must be fair, safe and ____.

(a) transparent ☐ **(b)** secret ☐ **(c)** costly ☐

B. Fill in the blanks. Use the word box.

behind **read** **phones** **errors**

1. Inclusive AI leaves no one ____.

2. Voice features help people who cannot ____.

3. Public AI should work on basic ____.

4. Citizens should be able to report ____.

C. Write T for True and F for False.

1. AI can support farmers in local languages. [ ]

2. Public AI only needs to work in English. [ ]

3. Accessibility includes features for people with disabilities. [ ]

4. Public AI should be checked for fairness. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Public good	____	**(a)** Usable by all abilities
2. Inclusive	____	**(b)** Benefits everyone
3. Accessible	____	**(c)** Open about how it works
4. Transparent	____	**(d)** Includes all groups

E. Answer in one word.

1. Which Indian mission builds language AI? _____

2. What word means usable by everyone? _____

3. AI for people, not just profit, is AI for public ____. _____

F. Answer these questions.

1. Give three examples of AI for public good in India.

2. Why must public AI be inclusive?

G. Case-based questions. Read and answer.

A farmer who couldn't read English used a Tamil voice AI app for crop advice. When one answer seemed wrong, a student checked with an officer and reported it. The app also worked on basic phones.

1. The app helped because it was in ____.

(a) the farmer's language ☐ **(b)** French ☐ **(c)** code ☐

2. Reporting the wrong answer improves ____.

(a) profits ☐ **(b)** accuracy and trust ☐ **(c)** battery life ☐

3. Working on basic phones makes it ____.

(a) slower ☐ **(b)** secret ☐ **(c)** accessible ☐

H. THINK DEEPER

Pick one public problem in your district. Design an AI-for-good solution and explain how you would make it inclusive.

UNIT REVIEW: AI SOLVE

Name: _____ Date: _____

Circle the right answer!

Q1. Long-term weather is...

a) Climate

b) Time

c) Tune

Q2. Nobody left out is...

a) Exclusive

b) Inclusive

c) Locked

Q3. Which is clean energy?

a) Oil

b) Coal

c) Solar

Q4. Bhashini helps with...

a) Indian languages

b) Games

c) Cooking

Q5. Watching Earth from space:

a) Balloon

b) Satellite

c) Kite

Q6. Good for the future is...

a) Wasteful

b) Polluting

c) Sustainable

WORD SEARCH: AI SOLVE

Find these words. They go across → or down ↓.

CLIMATE	SATELLITE	PUBLIC GOOD	INCLUSIVE	ACCESSIBLE	PLANET

I	K	D	C	R	R	C	M	C	O
E	A	K	O	O	D	B	W	W	P
I	N	C	L	U	S	I	V	E	L
B	N	U	L	T	I	S	H	B	A
K	A	C	D	W	T	B	G	O	N
K	W	I	E	B	L	L	M	E	E
N	N	C	L	I	M	A	T	E	T
P	S	A	T	E	L	L	I	T	E
P	U	B	L	I	C	G	O	O	D
A	C	C	E	S	S	I	B	L	E

I finished AI SOLVE! Teacher's sign: _____

MONTH 4

AI RISE

Skills • Careers • Future

Area: AI & Career

In this unit, you will explore:

Topic 7

Design Your AI Future

Topic 8

Skill-to-Career Map

TOPIC 7 • MONTH 4 • AI & CAREER

Design Your AI Future

LET'S LEARN

- Your future can be designed, like a project.
- Set SMART goals and build skills step by step.
- AI changes fast, so be a lifelong learner!

SMART goals

Specific

Measurable

Achievable

Time-bound

MAGIC WORDS

SMART goal
Clear, trackable goal

Mentor A guide

Lifelong Learning always

💬 TECH TALK: Design Your AI Future

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👧 **Riya**
Bolt, 'become successful' is my goal. Is that good 🤔?
10:02

🤖 **Bolt**
Too vague! Try a SMART goal 🎯: Specific, Measurable, Achievable, Relevant, Time-bound.
10:04 ✓✓

👦 **Kabir**
Example?
10:06

🤖 **Bolt**
'Finish a free Python course and build 2 projects by March.' Clear and checkable ✅.
10:08 ✓✓

👧 **Riya**
How do we reach big dreams?
10:10

🤖 **Bolt**
Design your future like a project 🗺️: vision goals skills weekly actions.
10:12 ✓✓

👦 **Kabir**

Who can guide us?

10:14

🤖 **Bolt**

A mentor 👩‍🏫: a teacher, senior or professional who shares experience and feedback.

10:16 ✓✓

👧 **Riya**

And since AI changes fast, we keep learning for life 📚!

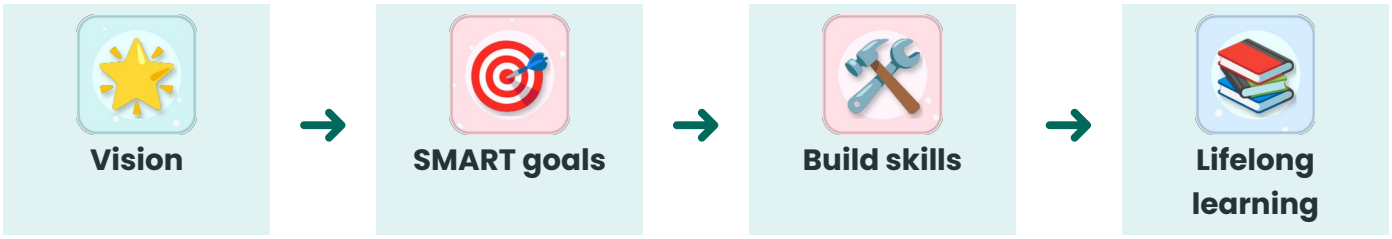
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Design your future like a project: set SMART goals, build skills step by step, find mentors and stay a lifelong learner.

🔄 How it works

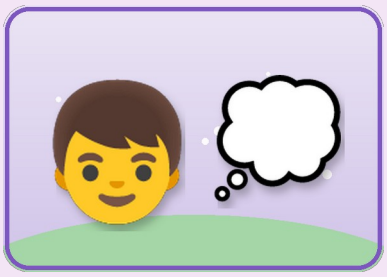


INDIA SPOTLIGHT

Free national platforms like SWAYAM offer online courses that students can use to keep learning beyond the classroom.

AI CHRONICLES: The Goal That Grew

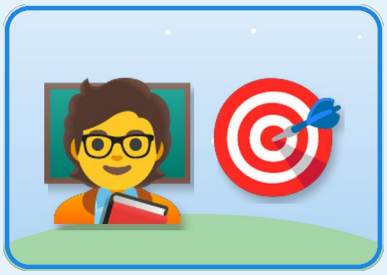
Kabir | Mentor Arjun Sir | Bolt



SCENE 1

Kabir's New Year goal was 'Become an AI expert.' By February, he'd done nothing 😞.

Arjun Sir, a former student now in tech, said: "Make it SMART. What exactly, by when?"



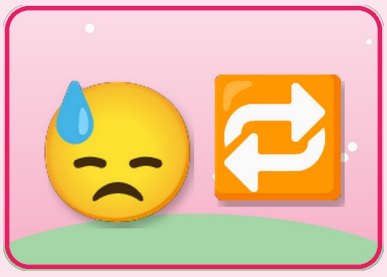
SCENE 2

SCENE 3

New goal: 'Finish a free Python course and build 2 AI projects by June.'

Kabir planned weekly actions: 3 lessons a week, a project every 6 weeks 💪.

SCENE 4



SCENE 5

In April he fell behind. Instead of quitting, he adjusted the plan and asked for feedback.

By June: course done, 2 projects shared. "Goals grow when you grow," smiled Arjun Sir.

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Kabir sets a SMART goal for maths	**Kabir goes back to vague goals**	**The goal runs away**

🧐 Story check: answer the questions.

1. Why did Kabir's first goal fail?

2. What made the new goal SMART?

3. What did Kabir do when he fell behind?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Future-ready habits

Learn one new skill each term

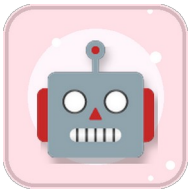
Try a free online course

Join olympiads & hackathons

Find a mentor

Build a portfolio

Follow tech news



BOLT SAYS:

"Learn Python" is a wish. "Finish 10 Python lessons by March" is a SMART goal!

FUN FACT!

Students in India can join coding contests, science fairs and olympiads to build their profile!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! My 1-Year Roadmap

You need:

Roadmap sheet

Colours

Calendar

Steps:

1. Write 1 big goal for this year.
2. Break it into 4 SMART mini-goals (one per term).
3. Draw a roadmap with milestones.
4. Share it with a mentor or parent.

THINK & TALK

What could stop you? How will you handle it?

Draw what you did!

WORKSHEET: Design Your AI Future (Part A)

Name: _____ Date: _____

1. SMART goal or vague wish? Tick ✓.

	SMART	Vague
Finish 10 Python lessons by March		
Be good at computers		
Run 2 km daily for 30 days		
Become famous		

2. Building your future. Write 1, 2, 3, 4.

Show your work ☐

Set a goal ☐

Learn skills ☐

Build projects ☐

3. Bonus: Draw your favourite picture from this topic!

I drew _____

WORKSHEET: Design Your AI Future (Part B)

Name: _____ Date: _____

1. Match the SMART letter to its meaning.

S	●	●	Time-bound
M	●	●	Specific
A	●	●	Achievable
T	●	●	Measurable

2. Draw your 1-year roadmap with 4 milestones!

My big goal: _____

PUZZLE FUN: Design Your AI Future

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

SMART goal

Meaning: _____

Example: _____

Mentor

Meaning: _____

Example: _____

Lifelong

Meaning: _____

Example: _____

🧪 DEMO LAB: Future Design Canvas (unplugged)

Name: _____ Date: _____

🎯 **Goal:** Design your next 3 years like a project.

🔗 **Open:** printed canvas / chart paper | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. Write a one-line vision: who you want to be in 10 years.
2. Turn it into 3 SMART goals for this year.
3. List 3 skills needed for each goal.
4. Plan weekly actions for the next 4 weeks.
5. Name one possible mentor and a question to ask them.

🔬 Record your observations

My vision:	
One SMART goal:	
My mentor question:	

💭 **Reflect:** Why is lifelong learning essential in an AI-driven world?

QUIZ TIME: Design Your AI Future

Circle the right answer!

Q1. The M in SMART is...

a) Measurable

b) Musical

c) Moony

Q2. A guide in your career is a...

a) Balloon

b) Mentor

c) Dinosaur

Q3. Learning always is...

a) Sleeping

b) Stopping

c) Lifelong learning

Q4. Which is a SMART goal?

a) 10 lessons by March

b) Be good

c) Get famous

Q5. AI changes...

a) Never

b) Fast

c) Not at all

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Design Your AI Future

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	SMART goals are specific and measurable.		
	"Be good at computers" is a SMART goal.		
	A mentor is a guide.		
	Lifelong learning means learning always.		
	AI never changes, so stop learning.		
	A portfolio shows your work.		

HOME CONNECT

Share your 1-year roadmap with your family. Ask them to be your accountability partners!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Design Your AI Future

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. The S in SMART goals stands for ____.

(a) Specific ☐ **(b)** Simple ☐ **(c)** Secret ☐

2. A guide who shares experience is a ____.

(a) mentee ☐ **(b)** mentor ☐ **(c)** monitor ☐

3. Learning throughout life is ____ learning.

(a) limited ☐ **(b)** lazy ☐ **(c)** lifelong ☐

4. Time-bound means the goal has a ____.

(a) deadline ☐ **(b)** colour ☐ **(c)** price ☐

B. Fill in the blanks. Use the word box.

Measurable actions learning future

1. M in SMART means ____.

2. Big dreams need weekly ____.

3. AI changes fast, so keep ____.

4. A vision is where you want to be in the ____.

C. Write T for True and F for False.

1. 'Be successful' is a SMART goal. [ ]

2. Mentors can give useful feedback. [ ]

3. Plans can be adjusted. [ ]

4. Learning ends after school. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Specific	____	**(a)** Realistic
2. Measurable	____	**(b)** Clear and exact
3. Achievable	____	**(c)** Has a deadline
4. Time-bound	____	**(d)** Can be checked

E. Answer in one word.

1. Who guides you with experience? _____

2. What kind of goal is clear and time-bound? _____

3. Learning for life is called? _____

F. Answer these questions.

1. Rewrite 'I want to be good at coding' as a SMART goal.

2. Why is lifelong learning important in the AI age?

G. Case-based questions. Read and answer.

Kabir's vague goal failed. With his mentor, he set a SMART goal: finish a Python course and build 2 projects by June. He fell behind in April but adjusted his plan and succeeded.

1. Kabir's first goal was too ____.

(a) vague ☐ **(b)** specific ☐ **(c)** short ☐

2. 'By June' makes the goal ____.

(a) achievable only ☐ **(b)** time-bound ☐ **(c)** secret ☐

3. Adjusting the plan shows ____.

(a) giving up ☐ **(b)** luck ☐ **(c)** resilience ☐

H. THINK DEEPER

Create a 1-year SMART plan to learn an AI skill. Include monthly milestones and how you'll measure progress.

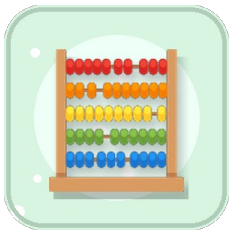
TOPIC 8 • MONTH 4 • AI & CAREER

Skill-to-Career Map

LET'S LEARN

- Every skill opens doors to many careers.
- Map your skills to roles in AI and beyond.
- Mix tech skills with people skills to stand out!

Skill families



Analytical

Technical

Creative

Interpersonal

MAGIC WORDS

Interdisciplinary
Mixing subjects

Upskill Learn new skills

T-shaped Deep + broad skills

💬 TECH TALK: Skill-to-Career Map

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir

Bolt, I'm good at maths and love talking to people 🗣️. Weird mix?

10:02

🤖 Bolt

A powerful mix! Many jobs need both: data analysts who explain findings, product managers 📱.

10:04 ✓✓

👧 Riya

What's a T-shaped person?

10:06

🤖 Bolt

Deep skill in one area (the |) plus broad skills across many (the —) 🔠.

10:08 ✓✓

👦 Kabir

And interdisciplinary?

10:10

🤖 Bolt

Combining fields: AI + medicine 🩺, AI + law ⚖️, AI + art 🎨.

10:12 ✓✓

👧 **Riya**

What if my skills become outdated?

10:14

🤖 **Bolt**

Upskill ⬆️: learn new skills regularly. AI changes jobs, so skills must grow.

10:16 ✓✓

👦 **Kabir**

Tech skills plus people skills = stand out ⭐!

10:18

😊 Type a message...

📌 **Pinned by Bolt**

Map your skills to many careers. Be T-shaped: deep in one area, broad in many. Combine fields and keep upskilling.

🔄 How it works

INDIA SPOTLIGHT

India's Skill India initiatives aim to train young people in future skills, including AI, data and digital technologies.

AI CHRONICLES: The T-Shaped Team

Riya

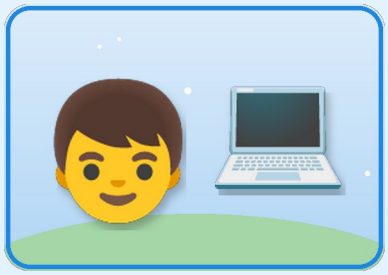
Kabir

Dr. Fatima

SCENE 1

A hospital challenge: 'Help patients understand their reports.' Riya's team signed up 🙋.

Kabir was deep in coding. Riya was deep in biology, and great at explaining 🗣️.


SCENE 2

SCENE 3

Dr. Fatima explained medical terms. The team realised: this needs AI + medicine + communication.

Kabir built a simple app; Riya wrote friendly explanations; another teammate designed the screens 🎨.

SCENE 4

SCENE 5

Kabir upskilled in basic medical terms; Riya learned basic Python. Both grew their T.

Their app won! Dr. Fatima said, "The future belongs to people who connect fields."

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
The team builds a medicine-reminder app	**They split up and work alone**	**The hospital becomes a robot**

🧐 Story check: answer the questions.

1. Which skills did each member bring?

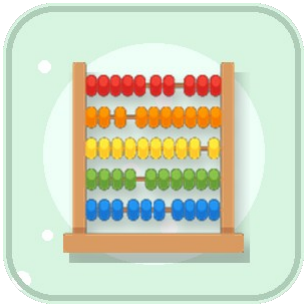
2. Why was the project interdisciplinary?

3. How did they upskill?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Skills → AI-age careers



Maths → data scientist

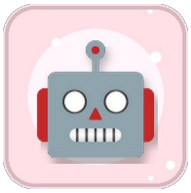
Coding → software engineer

Design → UX designer

Writing → tech journalist

Biology → bioinformatics

Ethics → AI policy



BOLT SAYS:

T-shaped skills: deep in one area, and broad across many!

FUN FACT!

Many new jobs mix subjects: like AI + medicine, or AI + agriculture!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Skill Map Gallery Walk

You need:

Big chart

Sticky notes

Class

Steps:

1		Write your 3 best skills on sticky notes.
2		Stick them on the class skill map.
3		Draw lines from skills to careers.
4		Gallery walk: find a surprising match!

THINK & TALK

Which career needs a skill you didn't expect?

Draw what you did!

WORKSHEET: Skill-to-Career Map (Part A)

Name: _____ Date: _____

1. Match the skill to an AI-age career.

Statistics ●

Design ●

Biology + code ●

Law + ethics ●

●

Bioinformatics

●

Data scientist

●

AI policy

●

UX designer

2. Fill in the blanks. Use the word box.

WORD BOX: upskill ✿ T-shaped ✿ interdisciplinary

Mixing subjects is _____.

To learn new skills is to _____.

Deep + broad skills are _____.

WORKSHEET: Skill-to-Career Map (Part B)

Name: _____ Date: _____

1. Tech skill or people skill? Tick ✓.

	Tech	People
Coding		
Teamwork		
Data analysis		
Public speaking		
Listening		

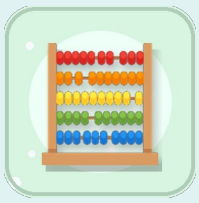
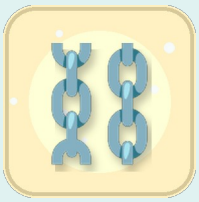
2. Draw your T-shaped skills: one deep skill + many broad ones!

My deep skill: _____

PUZZLE FUN: Skill-to-Career Map

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Interdisciplinary

Meaning: _____

Example: _____

Upskill

Meaning: _____

Example: _____

T-shaped

Meaning: _____

Example: _____

🧪 DEMO LAB: Skill-to-career mapping (online research with teacher)

Name: _____ Date: _____

🎯 **Goal:** Map your skills to roles and build a T-shaped plan.

🔗 **Open:** career portals + chart paper | 👩‍🏫 **With a teacher or parent**

👣 Steps

1. List 5 skills: 3 technical, 2 people skills.
2. Find 2 careers that use each skill.
3. Find one interdisciplinary role (e.g., AI + law).
4. Draw your T: deep skill on the stem, broad skills on the bar.
5. Choose one skill to upskill this term and a free course for it.

🔬 Record your observations

My deep skill:	
Interdisciplinary role found:	
Skill to upskill:	

💭 **Reflect:** Why do employers value T-shaped people?

QUIZ TIME: Skill-to-Career Map

Circle the right answer!

Q1. Mixing subjects is...

a) Interdisciplinary

b) Inflatable

c) Edible

Q2. Biology + coding can lead to...

a) Singing

b) Bioinformatics

c) Driving

Q3. Learning new skills is...

a) Downtime

b) Stopping

c) Upskilling

Q4. T-shaped skills are...

a) Deep + broad

b) Hidden

c) Only tall

Q5. Design skill can lead to...

a) Chemist

b) UX designer

c) Firefighter

MY SCORE: _____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: Skill-to-Career Map

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Interdisciplinary means mixing subjects.		
Biology + coding can lead to bioinformatics.		
Only one skill leads to one career.		
Upskilling means learning new skills.		
T-shaped skills are deep and broad.		
People skills don't matter in the AI age.		

HOME CONNECT

Ask family members which 3 skills helped them most at work. Add them to your map!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Skill-to-Career Map

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Deep in one skill and broad in many is ____.

(a) T-shaped ☐ **(b)** O-shaped ☐ **(c)** X-shaped ☐

2. Combining fields like AI + law is ____.

(a) outdated ☐ **(b)** interdisciplinary ☐ **(c)** illegal ☐

3. Learning new skills to stay relevant is to ____.

(a) downgrade ☐ **(b)** unplug ☐ **(c)** upskill ☐

4. Communication and teamwork are ____ skills.

(a) people ☐ **(b)** machine ☐ **(c)** hidden ☐

B. Fill in the blanks. Use the word box.

doors **field** **people** **India**

1. Every skill opens many ____.

2. AI + medicine is an interdisciplinary ____.

3. Mix tech skills with ____ skills.

4. India's Skill ____ trains future skills.

C. Write T for True and F for False.

1. One skill can lead to many careers. [ ]

2. T-shaped people know only one thing. [ ]

3. Upskilling helps when jobs change. [ ]

4. People skills don't matter in tech jobs. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. T-shaped	____	**(a)** Learn new skills
2. Interdisciplinary	____	**(b)** Deep + broad skills
3. Upskill	____	**(c)** Teamwork, communication
4. Soft skill	____	**(d)** Mixes fields

E. Answer in one word.

1. What shape describes deep + broad skills? _____

2. What is learning new skills called? _____

3. AI + art is what kind of field? _____

F. Answer these questions.

1. Map two of your skills to three careers each.

2. Explain the T-shaped model with your own example.

G. Case-based questions. Read and answer.

A team with a coder, a biology student who explained well and a designer built an app to explain medical reports. Each learned some of the others' skills.

1. The project was ____.

(a) interdisciplinary ☐ **(b)** single-subject ☐ **(c)** unplanned ☐

2. Learning each other's skills is ____.

(a) downgrading ☐ **(b)** upskilling ☐ **(c)** copying ☐

3. Their team shows the value of ____.

(a) working alone ☐ **(b)** only coding ☐ **(c)** combining skills ☐

H. THINK DEEPER

Which career might exist in 2040 that combines AI with your favourite subject? Describe it and the skills it needs.

UNIT REVIEW: AI RISE

Name: _____ Date: _____

Circle the right answer!

Q1. The T in SMART is...

a) Time-bound

b) Tree

c) Tune

Q2. Mixing subjects is...

a) Inflatable

b) Interdisciplinary

c) Edible

Q3. A career guide is a...

a) Dinosaur

b) Balloon

c) Mentor

Q4. Learning new skills is...

a) Upskilling

b) Downtime

c) Stopping

Q5. Which is a SMART goal?

a) Get famous

b) 20 lessons by June

c) Be good

Q6. Deep + broad skills are...

a) Hidden

b) Tall

c) T-shaped

WORD SEARCH: AI RISE

🔍 **Find these words. They go across → or down ↓.**

SMART GOAL	MENTOR	LIFELONG	UPSKILL	T-SHAPED	CAREER

K O T S H A P E D P
F S M A R T G O A L
H M E N T O R K I B
C B P I S T R L E G
C O G P I F M O U L
T G U P S K I L L L
D B H I C A R E E R
U W H D L F K P A B
M L I F E L O N G C
K L A L K L E O W C

I finished AI RISE! Teacher's sign: _____

MONTH 5

AI CREATE

Try • Build • Innovate

Area: AI Tools

In this unit, you will explore:

Topic 9

AI Research Room

Topic 10

AI Creator Studio

TOPIC 9 • MONTH 5 • AI TOOLS

AI Research Room

LET'S LEARN

- Research means finding, checking and using information.
- AI can summarise papers and suggest sources.
- Always verify sources and cite them properly!

Research steps

Ask a question

Search sources

Verify

Cite

MAGIC WORDS

Citation Credit to a source

Hallucination AI inventing facts

Hypothesis A testable guess

💬 TECH TALK: AI Research Room

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 Riya

Bolt, I'm researching plastic in rivers 🌊. Can AI do it for me?

10:02

🤖 Bolt

AI can help: summarise articles 📄, suggest keywords and explain hard terms.

10:04 ✓✓

👦 Kabir

Can we trust its sources?

10:06

🤖 Bolt

Not blindly! AI can hallucinate fake facts or even fake references ⚠️. Always open and check the source.

10:08 ✓✓

👧 Riya

How do we start real research?

10:10

🤖 Bolt

Ask a question, form a hypothesis 💡: 'Rivers near cities have more plastic.' Then gather evidence.

10:12 ✓✓

Kabir
And when we use someone's work?
10:14

Bolt
Cite it 📑: author, title, year and link. That's honest and lets readers check.
10:16 ✓✓

Riya
Question, search, verify, cite ✅!
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Research means finding, checking and using information. AI can summarise and suggest, but may hallucinate; verify every source and cite properly.

🔄 How it works

INDIA SPOTLIGHT
National digital libraries and open-access archives give Indian students free access to books and research papers.

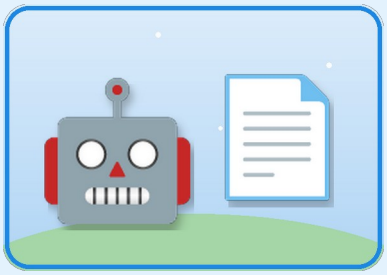
AI CHRONICLES: The Reference That Didn't Exist

Riya | Mr. Banerjee | Chatbot

SCENE 1

Riya's research project: 'Is plastic in our river increasing?' Her hypothesis: yes, near the city.

The chatbot gave her 3 impressive references with authors and years 📚.



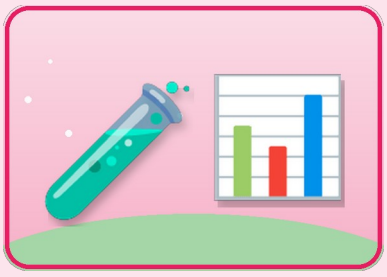
SCENE 2

SCENE 3

She searched each one. Two didn't exist! The AI had hallucinated them 😲.

She found real studies through Google Scholar and a pollution board report.

SCENE 4



SCENE 5

She collected water samples with her teacher. Evidence supported her hypothesis.

Mr. Banerjee gave full marks: "Your citations are real, and your evidence is honest."

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya presents at a science fair	**Riya uses fake references next time**	**The river writes a paper**

🧐 Story check: answer the questions.

1. What was Riya's hypothesis?

2. What problem did she find in the AI's references?

3. How did she find real sources?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Smart research with AI

Summarise a long article

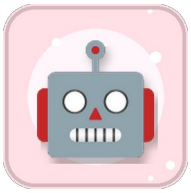
Find keywords to search

Explain a graph

Translate a source

Organise notes

Generate research questions



BOLT SAYS:
I can sometimes "hallucinate" and invent sources. Always check that a source really exists!

FUN FACT!
Citing sources gives credit to authors and lets readers check your facts!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Mini Research Project

You need:

Research question

Books & websites

Note template

Steps:

1

Pick a question: "Do plants grow faster with music?"

2

Write a hypothesis.

3

Find 3 sources. Verify each one.

4

Write a 1-page report with citations!

THINK & TALK

How did you check that each source was trustworthy?

Draw what you did!

WORKSHEET: AI Research Room (Part A)

Name: _____ Date: _____

1. Research steps. Write 1, 2, 3, 4.

Cite & write ☐

Ask a question ☐

Verify facts ☐

Search sources ☐

2. Trustworthy source or check carefully? Tick ✓.

	Trustworthy	Check carefully
Government data site		
Anonymous forward		
Published textbook		
AI answer without sources		
Science journal		

WORKSHEET: AI Research Room (Part B)

Name: _____ Date: _____

1. Match the research word to its meaning.

Citation ● ● AI inventing facts

Hypothesis ● ● Check if true

Hallucination ● ● Credit to a source

Verify ● ● A testable guess

2. Write your own research question and hypothesis!

My hypothesis: _____

PUZZLE FUN: AI Research Room

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Citation

Meaning: _____

Example: _____

Hallucination

Meaning: _____

Example: _____

Hypothesis

Meaning: _____

Example: _____

🧪 DEMO LAB: A school-approved AI chatbot + Google Scholar (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Use AI to start research, then verify and cite real sources.

🔗 **Open:** scholar.google.com | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Write a research question and a hypothesis.
2		Ask the chatbot for 3 sources on your topic.
3		Search each source on Google Scholar or the web. Does it exist?
4		Read one real source and summarise it in 5 lines.
5		Write a proper citation: author, title, year, link.

🔬 Record your observations

My hypothesis:	
How many AI sources were real?	
My citation:	

💭 **Reflect:** Why is checking AI-suggested references so important?

QUIZ TIME: AI Research Room

Circle the right answer!

Q1. Credit to a source is a...

a) Citation

b) Celebration

c) Cake

Q2. AI inventing facts is...

a) Harmony

b) Hallucination

c) Art

Q3. A testable guess is a...

a) Dice roll

b) Gift

c) Hypothesis

Q4. After finding a source, you...

a) Verify it

b) Ignore it

c) Forward it

Q5. Citing lets readers...

a) Sleep

b) Check facts

c) Play

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Research Room

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Citing gives credit to sources.		
	AI can invent sources that don't exist.		
	Every AI answer is automatically true.		
	A hypothesis is a testable guess.		
	Anonymous forwards are trustworthy sources.		
	Always verify before using a source.		

HOME CONNECT

Research a family question together (e.g., best time to water plants). Cite your sources!

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Research Room

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Giving credit to a source is a ____.

(a) citation ☐ **(b)** caption ☐ **(c)** conclusion ☐

2. When AI invents false facts, it is ____.

(a) citing ☐ **(b)** hallucinating ☐ **(c)** researching ☐

3. A testable prediction in research is a ____.

(a) headline ☐ **(b)** hashtag ☐ **(c)** hypothesis ☐

4. AI-suggested references must be ____.

(a) verified ☐ **(b)** copied ☐ **(c)** ignored ☐

B. Fill in the blanks. Use the word box.

question link Scholar support

1. Research starts with a ____.

2. A citation includes author, title, year and ____.

3. Find real papers on Google ____.

4. Evidence can ____ or reject a hypothesis.

C. Write T for True and F for False.

1. AI can create fake references. [ ]

2. Citing sources is optional in honest research. [ ]

3. A hypothesis can be tested. [ ]

4. AI summaries never need checking. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Hypothesis	____	**(a)** AI-made false info
2. Citation	____	**(b)** Testable prediction
3. Hallucination	____	**(c)** Facts that test a claim
4. Evidence	____	**(d)** Credit to a source

E. Answer in one word.

1. What is a testable prediction? _____

2. What is AI inventing facts called? _____

3. What gives credit to sources? _____

F. Answer these questions.

1. List the steps of a research process.

2. How can you check whether an AI-suggested source is real?

G. Case-based questions. Read and answer.

A chatbot gave Riya 3 references; 2 didn't exist. She found real studies on Google Scholar, collected samples and wrote correct citations.

1. The fake references were AI ____.

(a) hallucinations ☐ **(b)** citations ☐ **(c)** hypotheses ☐

2. She found real sources using ____.

(a) a game ☐ **(b)** Google Scholar ☐ **(c)** guessing ☐

3. Her samples provided ____.

(a) clickbait ☐ **(b)** opinions ☐ **(c)** evidence ☐

H. THINK DEEPER

Should students be allowed to use AI for research projects? Suggest rules that keep research honest.

TOPIC 10 • MONTH 5 • AI TOOLS

AI Creator Studio

LET'S LEARN

- Multimodal AI works with text, images, audio and video together.
- Creators combine tools to build projects quickly.
- Originality, honesty and credit make great creators!

Multimodal creating

Text

Image

Audio

Video

MAGIC WORDS

Multimodal Many media types

Original Your own creation

Infographic Data as pictures

💬 TECH TALK: AI Creator Studio

💬 **GAIO Squad** Riya, Kabir, Bolt • online

👦 **Kabir**
Bolt, our project needs a poster, a podcast and a short video 🎬!
10:02

🤖 **Bolt**
Multimodal AI can help: it works with text, images, audio 🎧 and video together.
10:04 ✓✓

👧 **Riya**
So one tool does everything?
10:06

🤖 **Bolt**
Some do! Others you combine: write a script, generate visuals, add voice, then edit ✂️.
10:08 ✓✓

👦 **Kabir**
What's an infographic?
10:10

🤖 **Bolt**
A visual that explains data quickly with icons, charts and short text 📊.
10:12 ✓✓

👧 **Riya**

How do we stay original 🌟?

10:14

🤖 **Bolt**

Start with your own idea, add your voice and research, and credit AI tools used 🏷️.

10:16 ✓✓

👦 **Kabir**

Original, honest, credited ✅!

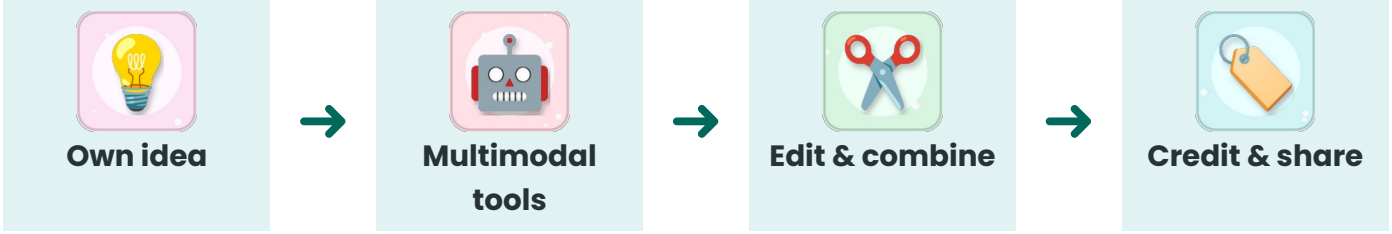
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Multimodal AI works with text, images, audio and video together. Great creators start with original ideas, combine tools and credit AI honestly.

🔄 How it works

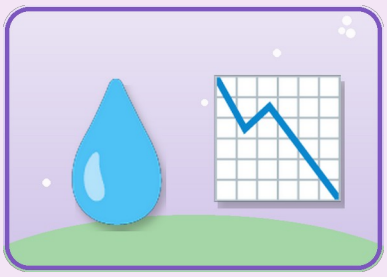


INDIA SPOTLIGHT

Many Indian classrooms now run digital storytelling and media projects where students create podcasts, videos and infographics.

AI CHRONICLES: The Water Week Campaign

Kabir Riya Principal



SCENE 1

The school's water tank emptied by noon. The principal asked for a 'Water Week' campaign 🚰.

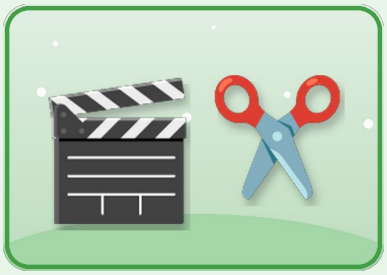
Riya made an infographic: litres used per tap, leaks found, and three simple tips.

SCENE 2

SCENE 3

Kabir wrote a podcast script in his own words, recorded it and used AI to clean the audio.

They used multimodal AI to turn photos and audio into a short video 📹.



SCENE 4

SCENE 5

The credits read: 'Ideas and research by students. AI tools used for audio cleanup and editing.'

Water use dropped by 25% in a month. The campaign was shared with 10 other schools 🎉!

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They create a campaign on electricity	**They copy another school's video**	**The tank starts a podcast**

🧐 Story check: answer the questions.

1. What did the infographic show?

2. How did they use multimodal AI?

3. How did they stay original and honest?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Student creator projects

School e-magazine

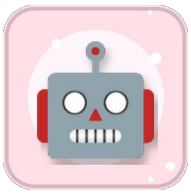
Science podcast

Explainer video

Infographic poster

Simple game

Illustrated story



BOLT SAYS:

Your unique idea is the star. I'm the backstage crew!

FUN FACT!

Infographics turn data into pictures so people understand faster!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Infographic Challenge

You need:

Class survey data

A3 paper

Colours

Steps:

1.
   Use your class survey data (or a new one).
2.
   Pick the 3 most interesting facts.
3.
   Design an infographic with icons and charts.
4.
   Add sources and credit any AI help.

THINK & TALK

Which design choice made your data easiest to understand?

Draw what you did!

WORKSHEET: AI Creator Studio (Part A)

Name: _____ Date: _____

1. Match the project to the main media type.

Podcast ●

E-magazine ●

Explainer ●

Infographic ●

●

Image

●

Audio

●

Text

●

Video

2. Sketch an infographic about AI in your school!

Title: _____

WORKSHEET: AI Creator Studio (Part B)

Name: _____ Date: _____

1. Making a multimodal project. Write 1, 2, 3, 4.

Credit & share ☐

Plan the message ☐

Create media ☐

Review & improve ☐

2. Fill in the blanks. Use the word box.

WORD BOX: multimodal ✿ infographic ✿ original

Many media types together is _____.

Data as pictures is an _____.

My own creation is _____.

3. Bonus: Draw your favourite picture from this topic!

I drew _____

PUZZLE FUN: AI Creator Studio

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Multimodal

Meaning: _____

Example: _____

Original

Meaning: _____

Example: _____

Infographic

Meaning: _____

Example: _____

🧪 DEMO LAB: Canva for Education + any AI audio tool (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Create a multimodal mini-project: infographic + 1-minute audio.

🔗 **Open:** canva.com/education | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Pick a topic: 'Water use in my home'.
2		Collect 3 facts and design an infographic with icons and a chart.
3		Write a 1-minute script in your own words.
4		Record it or use AI voice, then clean the audio.
5		Add a credits panel listing all AI tools used.

🔬 Record your observations

Facts used:	
AI tools used:	
What I created myself:	

💭 **Reflect:** What makes a creation original even when AI tools help?

QUIZ TIME: AI Creator Studio

Circle the right answer!

Q1. Many media types together is...

a) Multimodal

b) Muted

c) Inflated

Q2. Data shown as pictures is an...

a) Ice-cream

b) Infographic

c) Interview

Q3. Your own creation is...

a) Copied

b) Deleted

c) Original

Q4. A podcast is mostly...

a) Audio

b) Images

c) Food

Q5. Great creators give...

a) Nothing

b) Credit

c) Lies

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Creator Studio

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Multimodal AI uses text, images, audio and video.		
Infographics turn data into pictures.		
Copying is the same as being original.		
Great creators credit AI help.		
A podcast is mostly video.		
The idea is the star of a project.		

HOME CONNECT

Create a family infographic: our week in numbers (sleep, steps, screen time).

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: AI Creator Studio

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. AI that handles text, images and audio together is ____.

(a) multimodal ☐ **(b)** monomodal ☐ **(c)** manual ☐

2. A visual that explains data with icons and charts is an ____.

(a) invoice ☐ **(b)** infographic ☐ **(c)** index ☐

3. Starting with your own idea keeps work ____.

(a) copied ☐ **(b)** random ☐ **(c)** original ☐

4. Listing AI tools used is giving ____.

(a) credit ☐ **(b)** trouble ☐ **(c)** orders ☐

B. Fill in the blanks. Use the word box.

video **words** **data** **credit**

1. Multimodal AI works with text, images, audio and ____.

2. Write the script in your own ____.

3. Infographics explain ____ quickly.

4. Honest creators ____ AI tools.

C. Write T for True and F for False.

1. Infographics use visuals to explain data. [ ]

2. Using AI means the work cannot be original. [ ]

3. Crediting AI tools is honest. [ ]

4. Multimodal AI only works with text. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Multimodal	____	(a) Your own idea
2. Infographic	____	(b) Many media types
3. Original	____	(c) Tools and people acknowledged
4. Credits	____	(d) Visual data story

E. Answer in one word.

1. What is AI that uses many media types? _____
2. What visual explains data with icons? _____
3. What do we add to acknowledge tools? _____

F. Answer these questions.

1. Plan a multimodal project on a social issue.

2. How can AI-assisted work still be original?

G. Case-based questions. Read and answer.

Students made an infographic on water use, a podcast in their own words with AI-cleaned audio and an AI-assisted video. Credits listed the AI tools. Water use dropped 25%.

1. The project was ____.

(a) multimodal ☐ **(b)** text-only ☐ **(c)** copied ☐

2. The podcast script was ____.

(a) AI-written ☐ **(b)** their own words ☐ **(c)** plagiarised ☐

3. The credits show ____.

(a) secrecy ☐ **(b)** laziness ☐ **(c)** honesty ☐

H. THINK DEEPER

Design a multimodal campaign for road safety. Which parts would you create yourself and where would AI help?

UNIT REVIEW: AI CREATE

Name: _____ Date: _____

Circle the right answer!

Q1. Credit to a source is a...

a) Citation

b) Celebration

c) Cake

Q2. Many media types together:

a) Muted

b) Multimodal

c) Inflated

Q3. AI inventing facts is...

a) Harmony

b) Art

c) Hallucination

Q4. Data as pictures is an...

a) Infographic

b) Ice-cream

c) Interview

Q5. A testable guess is a...

a) Dice roll

b) Hypothesis

c) Gift

Q6. Your own creation is...

a) Copied

b) Deleted

c) Original

WORD SEARCH: AI CREATE

🔍 **Find these words. They go across → or down ↓.**

CITATION

HYPOTHESIS

MULTIMODAL

ORIGINAL

SOURCE

VERIFY

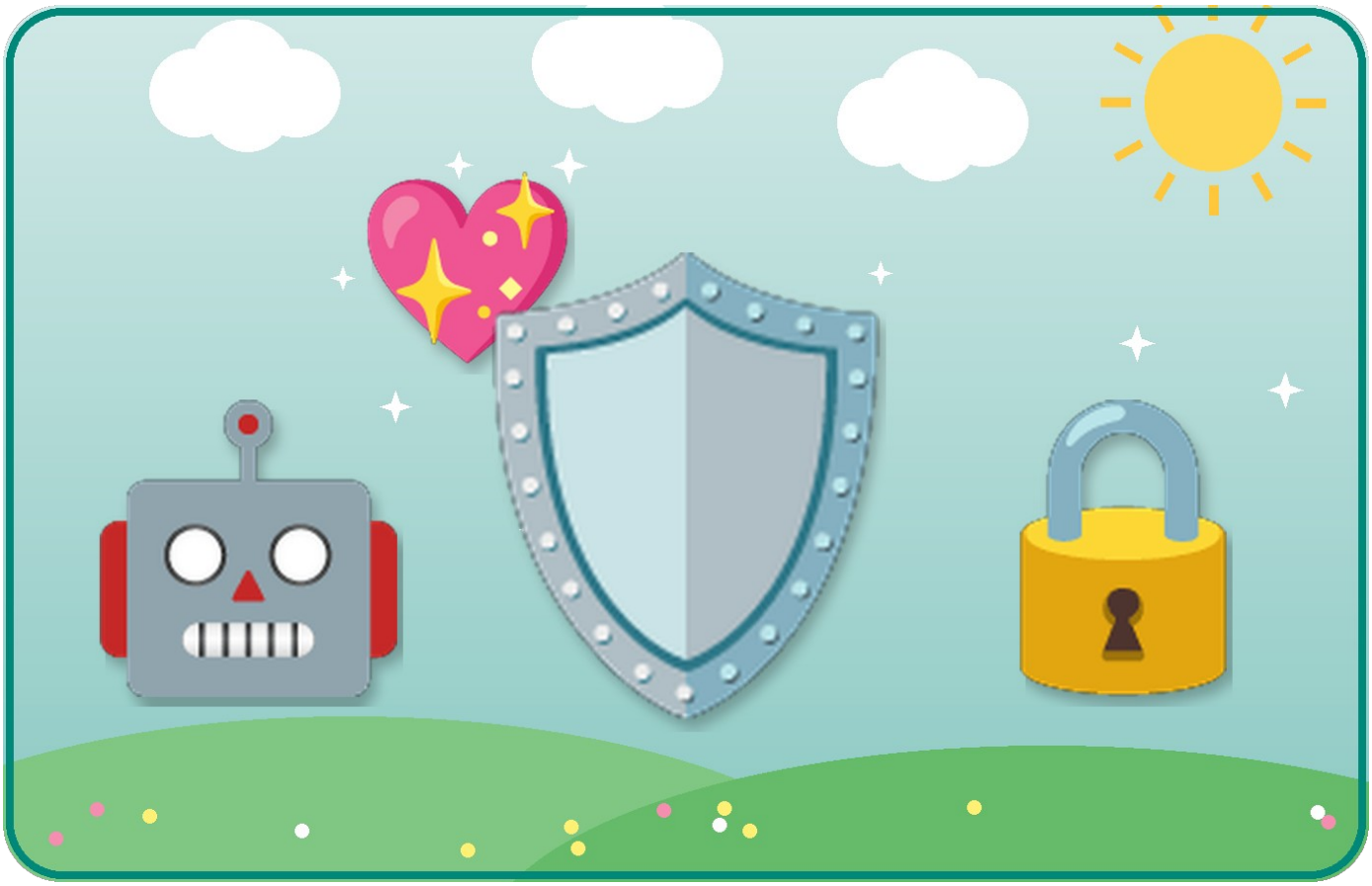
U	M	C	A	P	B	F	W	H	G
D	U	I	H	P	M	S	M	Y	S
I	L	T	O	P	D	V	U	P	M
K	T	A	R	B	O	E	C	O	G
L	I	T	I	S	S	R	W	T	M
E	M	I	G	L	O	I	I	H	T
C	O	O	I	K	U	F	L	E	K
F	D	N	N	C	R	Y	E	S	K
R	A	F	A	B	C	C	W	I	T
N	L	B	L	H	E	W	M	S	I

I finished AI CREATE! Teacher's sign: _____

MONTH 6

AI CARE

Verify • Protect • Respect



Area: Responsible AI

In this unit, you will explore:

Topic 11

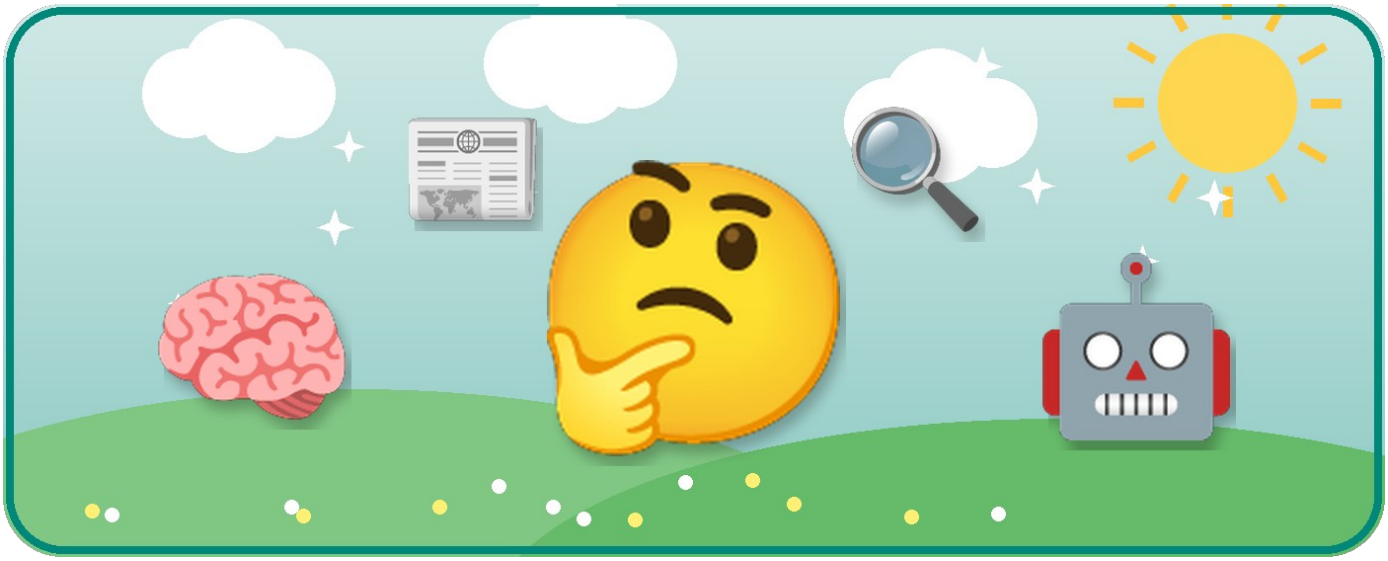
Think Before You Believe

Topic 12

Digital Footprints & AI

TOPIC 11 • MONTH 6 • RESPONSIBLE AI

Think Before You Believe



LET'S LEARN

- Our brains have shortcuts called biases.
- Apps can show only what we like: a filter bubble.
- Critical thinkers question, compare and stay open-minded!

Thinking traps

Filter bubble

Confirmation bias

Emotional bait

AI hallucination

MAGIC WORDS

Confirmation bias Believing what we like

Filter bubble Seeing one side only

Clickbait Tricky headline

💬 TECH TALK: Think Before You Believe

💬 GAIO Squad Riya, Kabir, Bolt • online

👧 **Riya**

Bolt, my feed says only one phone brand is good 📱. Everyone agrees!

10:02

🤖 **Bolt**

Maybe you're in a filter bubble 🫧: apps show you more of what you already like.

10:04 ✓✓

👦 **Kabir**

Why does that matter?

10:06

🤖 **Bolt**

You stop seeing other views. Add confirmation bias 🧠: we believe what matches our opinion.

10:08 ✓✓

👧 **Riya**

What's clickbait?

10:10

🤖 **Bolt**

Shocking headlines made to get clicks 🎣, like 'You won't BELIEVE what happened!'

10:12 ✓✓

Kabir
How do we think critically?
10:14

Bolt
Question the claim ❓, compare sources, look for evidence and stay open to changing your mind.
10:16 ✓✓

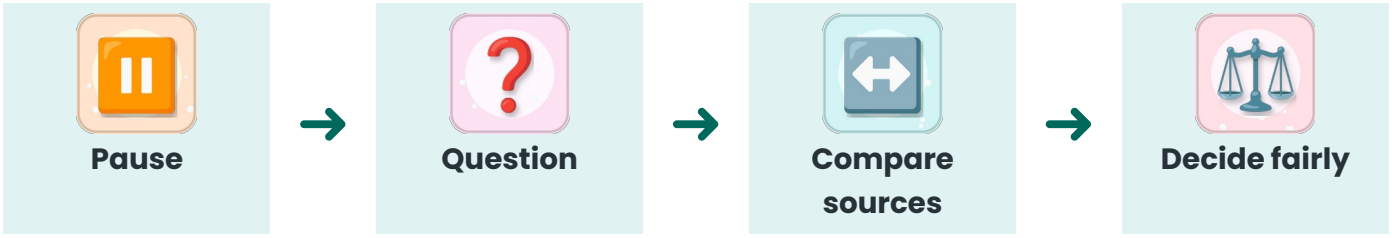
Riya
Pause, question, compare, decide ⚖️!
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Confirmation bias makes us favour what we already believe; filter bubbles show us only what we like; clickbait tricks us. Critical thinkers question, compare and stay open-minded.

🔄 How it works



INDIA SPOTLIGHT

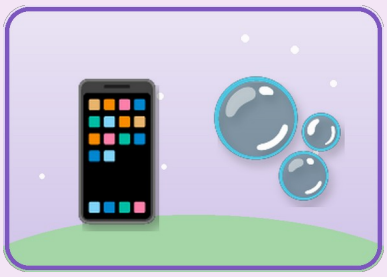
Media literacy programmes in Indian schools teach students to recognise misinformation and verify claims before sharing.

AI CHRONICLES: Inside the Bubble

Kabir

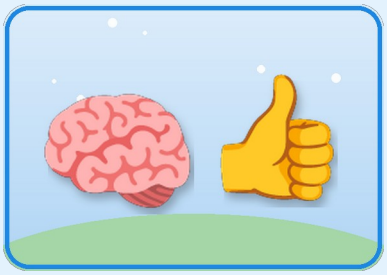
Riya

The Feed


SCENE 1

Kabir's feed showed only videos saying a new energy drink was 'healthy' 🥤.

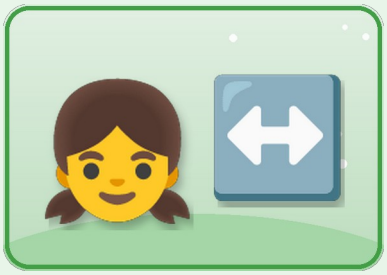
He liked it already, so every video felt true. That's confirmation bias!

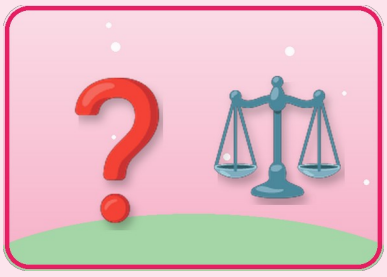

SCENE 2

SCENE 3

One headline screamed: 'Doctors HATE this drink secret!' Pure clickbait 🎣.

Riya compared sources: a health ministry advisory and two reliable news sites warned about high caffeine.


SCENE 4


SCENE 5

Kabir questioned his belief, read the evidence and changed his mind 🔄.

They made a 'Pop Your Bubble' poster: "Pause, question, compare, decide!"

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
They run a media literacy workshop	**Kabir buys ten more drinks**	**The bubble pops itself**

🧐 Story check: answer the questions.

1. What is a filter bubble?

2. How did confirmation bias affect Kabir?

3. How did Riya check the claim?

Rate this story: shade the stars.

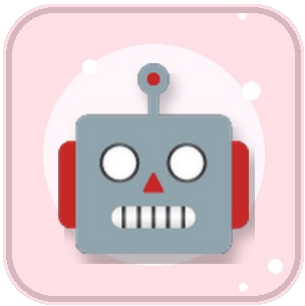
☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Where thinking traps hide

Endless video feeds

Clickbait headlines

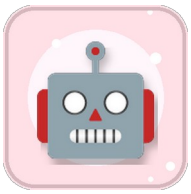
Group-chat rumours



Confident AI mistakes

Misleading graphs

Fake reviews



BOLT SAYS:

I can sound very confident and still be wrong. Confidence is not proof!

FUN FACT!

Recommendation algorithms learn what keeps you watching, which can create filter bubbles!

I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Burst the Bubble

You need:

Two news articles

Compare chart

Pairs

Steps:

1. Read two articles on the same event.
2. List what's the same and what's different.
3. Spot emotional words and missing facts.
4. Write a balanced 3-line summary.

THINK & TALK

Why do we like information that agrees with us?

Draw what you did!

WORKSHEET: Think Before You Believe (Part A)

Name: _____ Date: _____

1. Match the thinking trap to the example.

Filter bubble ●

Confirmation bias ●

Clickbait ●

Hallucination ●

● "You won't BELIEVE this!"

● Feed shows only one side

● AI invents a fake source

● Ignoring facts you dislike

2. Fill in the blanks. Use the word box.

WORD BOX: clickbait ✿ bubble ✿ bias

A tricky headline is _____.

Seeing one side only is a filter _____.

Believing only what we like is confirmation _____.

WORKSHEET: Think Before You Believe (Part B)

Name: _____ Date: _____

1. Circle the signs of clickbait.

SHOCKING!!!

Date & author

You won't believe...

Cited sources

Click to find out!

ALL CAPS anger

2. Critical thinker or not? Tick ✓.

		Critical	Not critical
	Compares two sources		
	Shares because it agrees with me		
	Asks "Who made this?"		
	Trusts AI without checking		

PUZZLE FUN: Think Before You Believe

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Confirmation bias

Meaning: _____

Example: _____

Filter bubble

Meaning: _____

Example: _____

Clickbait

Meaning: _____

Example: _____

🧪 DEMO LAB: Two news sites + feed experiment (with teacher)

Name: _____ Date: _____

🎯 **Goal:** Spot bias, filter bubbles and clickbait.

🔗 **Open:** any two reliable news websites | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Pick one news event. Read it on two different reliable sites.
2		List differences in headline, facts and tone.
3		Collect 3 clickbait headlines and rewrite them honestly.
4		Discuss: what does your video app usually show you? Why?
5		Write one belief you hold and one fact that challenges it.

🔬 Record your observations

Differences between sites:	
One honest rewrite:	
A bubble I'm in:	

💭 **Reflect:** How can we break out of filter bubbles?

? QUIZ TIME: Think Before You Believe

Circle the right answer!

Q1. Seeing only one side is a...

a) Filter bubble

b) Balloon

c) Bath

Q2. Believing what we already like is...

a) Accuracy

b) Confirmation bias

c) Rhythm

Q3. A confident AI answer is...

a) Always true

b) A party

c) Not proof

Q4. "You won't BELIEVE this!" is...

a) Clickbait

b) Research

c) Proof

Q5. Critical thinkers...

a) Believe everything

b) Question & compare

c) Ignore all

MY SCORE: ____ / 5. Shade one star per correct answer.

✅ TRUE OR FALSE: Think Before You Believe

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	A filter bubble shows us one side.		
	Confirmation bias means believing what we already like.		
	A confident AI answer is always correct.		
	Clickbait uses tricky headlines.		
	Graphs can never mislead.		
	Critical thinkers compare sources.		

HOME CONNECT

Watch one news story with your family from two sources. What was different?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Think Before You Believe

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. Favouring information that matches our beliefs is ____ bias.

(a) confirmation ☐ **(b)** conversion ☐ **(c)** connection ☐

2. Seeing only content you already like is a filter ____.

(a) paper ☐ **(b)** bubble ☐ **(c)** box ☐

3. Shocking headlines made for clicks are ____.

(a) citations ☐ **(b)** editorials ☐ **(c)** clickbait ☐

4. Critical thinkers ____ sources.

(a) compare ☐ **(b)** ignore ☐ **(c)** copy ☐

B. Fill in the blanks. Use the word box.

biases **like** **open** **decide**

1. Our brain's mental shortcuts are ____.

2. Apps show more of what we ____.

3. Stay ____-minded.

4. Pause, question, compare, ____.

C. Write T for True and F for False.

1. Filter bubbles can hide other viewpoints. [ ]

2. Clickbait headlines are always accurate. [ ]

3. Changing your mind with evidence is good thinking. [ ]

4. Confirmation bias affects only other people. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Confirmation bias	____	**(a)** Tricky headline
2. Filter bubble	____	**(b)** Favour own beliefs
3. Clickbait	____	**(c)** Question and compare
4. Critical thinking	____	**(d)** Only similar content

E. Answer in one word.

1. What tricky headlines get clicks? _____

2. What hides other views on feeds? _____

3. What bias favours our beliefs? _____

F. Answer these questions.

1. Explain filter bubbles and how algorithms create them.

2. Give three steps to check a viral claim.

G. Case-based questions. Read and answer.

Kabir's feed showed only videos praising an energy drink. He believed them because he already liked it. Riya compared reliable sources warning about high caffeine, and Kabir changed his mind.

1. Kabir's feed was a _____.

(a) filter bubble ☐ **(b)** library ☐ **(c)** textbook ☐

2. Believing because he liked it shows _____.

(a) fairness ☐ **(b)** confirmation bias ☐ **(c)** research ☐

3. Changing his mind with evidence shows _____.

(a) weakness ☐ **(b)** bias ☐ **(c)** critical thinking ☐

H. THINK DEEPER

Should social media apps be required to show users different viewpoints? Argue both sides.

TOPIC 12 • MONTH 6 • RESPONSIBLE AI

Digital Footprints & AI

LET'S LEARN

- Your digital footprint is the trail you leave online.
- AI can analyse footprints to profile and target people.
- Think before you post: the internet rarely forgets!

Your footprint includes

Posts & photos

Searches

Likes

Locations

MAGIC WORDS

Footprint Online trail

Profiling Guessing who you are

Reputation How others see you

💬 TECH TALK: Digital Footprints & AI

💬 GAIO Squad Riya, Kabir, Bolt • online

👦 Kabir

Bolt, I searched for sneakers once 👟. Now ads follow me everywhere!

10:02

🤖 Bolt

That's your digital footprint 👣: searches, likes, posts and locations leave a trail.

10:04 ✓✓

👧 Riya

How do companies use it?

10:06

🤖 Bolt

AI analyses footprints to profile you 🧩: age, interests, habits. Then it targets ads.

10:08 ✓✓

👦 Kabir

Can footprints affect my future?

10:10

🤖 Bolt

Yes. Colleges and employers may look online 🔍. Your reputation is built by what you post.

10:12 ✓✓

Riya
Can we delete everything?
10:14

Bolt
Not always. Screenshots and copies spread 📸. The internet rarely forgets.
10:16 ✓✓

Kabir
Think before you post, check privacy settings 🔒!
10:18

😊 Type a message...

📌 **Pinned by Bolt**

Your digital footprint is the trail you leave online. AI can profile and target you. Think before you post and manage privacy to protect your reputation.

🔄 How it works

INDIA SPOTLIGHT
India's data protection law gives people rights to access and correct their personal data and to withdraw consent.

AI CHRONICLES: The Post That Came Back

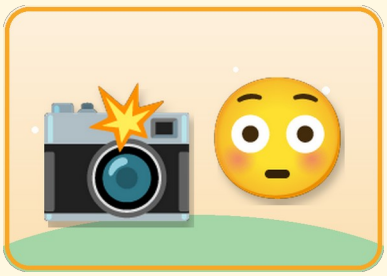
Riya | Senior Rohan | Interview Panel

SCENE 1

In Class 9, Rohan posted a rude meme mocking a classmate. "Just a joke," he said 😏.

Years later, he applied for a top internship. The panel searched his name online 🔍.

SCENE 2



SCENE 3

The old meme appeared in a screenshot. He had deleted it, but copies remained.

He was asked to explain. He apologised honestly, but lost the chance 😔.

SCENE 4

SCENE 5

Hearing this, Riya checked her privacy settings and cleaned up her old posts.

She started posting her projects instead. "Let your footprint show your best self."

SCENE 6

🔮 What happens next? Tick ✓ one!

☐	☐	☐
Riya builds a positive online portfolio	**Riya posts without thinking**	**The internet forgets everything**

🧐 Story check: answer the questions.

1. Why did Rohan's old post matter later?

2. Why couldn't deleting it fully help?

3. What did Riya change?

Rate this story: shade the stars.

☆ ☆ ☆ ☆ ☆

LOOK AROUND YOU: Footprint smart habits

Private account settings

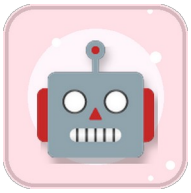
Pause before posting

Delete old risky posts

Don't post live location

Search your own name

Build a positive footprint



BOLT SAYS:
Colleges and employers may look online. Make your footprint something you're proud of!

FUN FACT!
Ad systems use AI to guess your interests from what you watch, like and search!

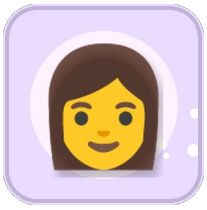
I spy AI! Draw one AI helper you saw today.

LET'S DO IT! Footprint Audit

You need:

Audit checklist

Device (with adult)



A parent

Steps:

1		With a parent, search your name online.
2		List what you find: photos, posts, mentions.
3		Check privacy settings on one account.
4		Plan 2 actions for a positive footprint.

THINK & TALK

What would you want a future college to see about you?

Draw what you did!

WORKSHEET: Digital Footprints & AI (Part A)

Name: _____ Date: _____

1. Smart post or risky post? Tick ✓.

	Smart	Risky
Science fair certificate		
Live location at home		
Artwork you made		
Angry rant about a classmate		
Photo of exam hall ticket		

2. Fill in the blanks. Use the word box.

WORD BOX: footprint ✿ reputation ✿ profiling

My online trail is my digital _____.

AI guessing who I am is _____.

How others see me is my _____.

WORKSHEET: Digital Footprints & AI (Part B)

Name: _____ Date: _____

1. Match the footprint to the smart action.

	Live location post	●	Delete it
	Angry old comment	●	Share it proudly
	Public profile	●	Turn it off
	Achievement	●	Make it private

2. Design your ideal positive online profile!

My profile headline: _____

PUZZLE FUN: Digital Footprints & AI

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Footprint

Meaning: _____

Example: _____

Profiling

Meaning: _____

Example: _____

Reputation

Meaning: _____

Example: _____

🧪 DEMO LAB: Search engine + account privacy dashboards (with a parent/teacher)

Name: _____ Date: _____

🎯 **Goal:** Audit your digital footprint and improve privacy.

🔗 **Open:** your Google/app account privacy settings | 👩‍🏫 **With a teacher or parent**

👣 Steps

1		Search your own name (with a parent). Note what appears.
2		List 5 apps that hold your data.
3		Open an account's 'ad personalisation' or activity settings.
4		Turn off unnecessary tracking and set profiles to private.
5		Write 3 rules for posting online.

🔬 Record your observations

What I found about me:	
Settings changed:	
My posting rules:	

💭 **Reflect:** How could your digital footprint help or harm you in 5 years?

? QUIZ TIME: Digital Footprints & AI

Circle the right answer!

Q1. Your online trail is a...

a) Digital footprint

b) Real foot

c) Map

Q2. AI guessing your interests is...

a) Painting

b) Profiling

c) Singing

Q3. Before posting, you should...

a) Post fast

b) Rant

c) Pause & think

Q4. Posting live home location is...

a) Risky

b) Smart

c) Fun

Q5. The internet usually...

a) Forgets fast

b) Remembers

c) Sleeps

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Digital Footprints & AI

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	A digital footprint is your online trail.		
	The internet forgets everything quickly.		
	Ad AI can profile your interests.		
	Posting live home location is risky.		
	You can build a positive footprint.		
	Angry rants improve your reputation.		

HOME CONNECT

With a parent, check the privacy settings on one family account together.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

EXERCISE: Digital Footprints & AI

Name: _____ Date: _____

A. Tick (✓) the correct answer.

1. The trail you leave online is your digital ____.

(a) footprint ☐ **(b)** fingerprint ☐ **(c)** photograph ☐

2. AI building a picture of your interests is ____.

(a) printing ☐ **(b)** profiling ☐ **(c)** posting ☐

3. What others think of you online is your ____.

(a) router ☐ **(b)** resolution ☐ **(c)** reputation ☐

4. Deleted posts may survive as ____.

(a) screenshots ☐ **(b)** passwords ☐ **(c)** batteries ☐

B. Fill in the blanks. Use the word box.

post **ads** **private** **forgets**

1. Think before you ____.

2. Targeted ____ use your profile.

3. Set profiles to ____.

4. The internet rarely ____.

C. Write T for True and F for False.

1. Searches and likes add to your footprint. [ ]

2. Deleting a post removes every copy forever. [ ]

3. Employers may check online profiles. [ ]

4. Privacy settings are useless. [ ]

D. Match the following. Write the letter.

Column A		Column B
1. Footprint	____	**(a)** How others see you
2. Profiling	____	**(b)** Online trail
3. Reputation	____	**(c)** Control who sees data
4. Privacy settings	____	**(d)** Building a picture of you

E. Answer in one word.

1. What is your online trail? _____

2. What does AI do with footprints for ads? _____

3. What protects who sees your posts? _____

F. Answer these questions.

1. List five things that add to your digital footprint.

2. How can a positive footprint help your future?

G. Case-based questions. Read and answer.

Rohan posted a rude meme in Class 9 and deleted it later. Years afterwards, a screenshot appeared when an internship panel searched his name, and he lost the chance.

1. The meme became part of his ____.

(a) digital footprint ☐ **(b)** textbook ☐ **(c)** password ☐

2. Deleting didn't help because of ____.

(a) Wi-Fi ☐ **(b)** screenshots ☐ **(c)** batteries ☐

3. The incident hurt his ____.

(a) phone ☐ **(b)** router ☐ **(c)** reputation ☐

H. THINK DEEPER

Write a 'digital footprint policy' for your class: five rules that protect privacy and build positive reputations.

UNIT REVIEW: AI CARE

Name: _____ Date: _____

Circle the right answer!

Q1. Seeing only one side is a...

a) Filter bubble

b) Balloon

c) Bath

Q2. Your online trail is a...

a) Map

b) Digital footprint

c) Real foot

Q3. "You won't BELIEVE this!" is...

a) Research

b) Proof

c) Clickbait

Q4. AI guessing your interests is...

a) Profiling

b) Painting

c) Singing

Q5. Confident AI answers are...

a) Always true

b) Not proof

c) Parties

Q6. Before posting, you...

a) Post fast

b) Rant

c) Pause & think

WORD SEARCH: AI CARE

Find these words. They go across → or down ↓.

CLICKBAIT

FOOTPRINT

PROFILING

REPUTATION

BIAS

PRIVACY

C	L	C	M	O	I	P	R	D	G
L	K	D	B	U	F	G	E	P	M
I	R	G	S	U	O	S	P	R	P
C	A	M	H	W	O	O	U	O	R
K	K	M	U	D	T	C	T	F	I
B	S	S	G	D	P	W	A	I	V
A	I	K	G	N	R	B	T	L	A
I	R	H	E	W	I	I	I	I	C
T	G	S	A	G	N	A	O	N	Y
F	A	L	T	W	T	S	N	G	W

I finished AI CARE! Teacher's sign: _____

MY PICTURE DICTIONARY (1/2)

Abstraction
Keep only what matters

Accessible
Usable by all

Accuracy
How often right

Citation
Credit to a source

Classification
Predicts a group

Clickbait
Tricky headline

Climate
Long-term weather

Confirmation bias
Believing what we like

Dictionary
Key-value pairs

Efficient
Fast, less waste

Filter bubble
Seeing one side only

Footprint
Online trail

Hallucination
AI inventing facts

Hypothesis
A testable guess

Inclusive
Nobody left out

Infographic
Data as pictures

Information
Organised data

Insight
Useful discovery

MY PICTURE DICTIONARY (2/2)

Interdisciplinary
Mixing subjects

Lifelong
Learning always

Mentor
A guide

Multimodal
Many media types

Original
Your own creation

Parameter
Input to a function

Profiling
Guessing who you are

Public good
Benefits everyone

Raw data
Unprocessed facts

Regression
Predicts a number

Reputation
How others see you

Return
Send back a result

Satellite
Watches from space

SMART goal
Clear, trackable goal

Sustainable
Good for the future

T-shaped
Deep + broad skills

Test case
Input to check code

Upskill
Learn new skills

? OLYMPIAD PRACTICE TEST • Part 1 of 4

Read each question. Circle the right answer. 20 questions in all!

Q1. "38°, 39°, 41°" is an example of...

a) Data

b) Wisdom

c) Music

Q2. "Heatwave coming, stay hydrated" is...

a) Raw data

b) Wisdom/intelligence

c) Noise

Q3. Predicting tomorrow's temperature is...

a) Classification

b) Art

c) Regression

Q4. A model right 90 times out of 100 has...

a) 90% accuracy

b) 100% accuracy

c) 10% accuracy

Q5. d = {"a": 5}; d["a"] gives...

a) a

b) 5

c) Error

? OLYMPIAD PRACTICE TEST • Part 2 of 4

Q6. def add(a, b): return a + b → add(2, 3) is...

a) 23

b) 6

c) 5

Q7. A metro map ignoring trees shows...

a) Abstraction

b) Decomposition

c) A bug

Q8. Testing 0, 50 and 100 marks checks...

a) Balloons

b) Edge cases

c) Pizza

Q9. Satellites help AI to track...

a) Songs

b) Games

c) Forests

Q10. Which is NOT clean energy?

a) Coal

b) Solar

c) Wind

? OLYMPIAD PRACTICE TEST • Part 3 of 4

Q11. Crop advice in Tamil by voice is AI for...

a) Only profit

b) Public good

c) Gaming

Q12. "Finish 20 lessons by June" is a...

a) Vague wish

b) Toy

c) SMART goal

Q13. Biology + coding = ...

a) Bioinformatics

b) Singing

c) Farming only

Q14. An AI-suggested source that doesn't exist is a...

a) Citation

b) Hallucination

c) Hypothesis

Q15. Text + image + audio together is...

a) Silent

b) Single-media

c) Multimodal

? OLYMPIAD PRACTICE TEST • Part 4 of 4

Q16. Infographics show data as...

a) Pictures

b) Songs

c) Food

Q17. Sharing only what agrees with you is...

a) Research

b) Confirmation bias

c) Fairness

Q18. Feeds showing one side create a...

a) Rainbow

b) Wall

c) Filter bubble

Q19. Posting your exam hall ticket online is...

a) Risky

b) Smart

c) Fun

Q20. A positive footprint shows your...

a) Anger

b) Achievements

c) Live location

MY SCORE: I got ____ / 20. Colour a star for every 4 right!

CERTIFICATE

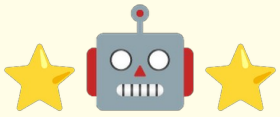
of AI Explorer

This is to say that

_____

of Class 9 finished the 6-month AI journey:

DISCOVER → CONNECT → SOLVE → RISE → CREATE → CARE



Teacher: _____ Date: _____

? ANSWER KEY (for teachers & parents)

1. From Data to Intelligence	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 1-Data, 2-Information, 3-Knowledge, 4-Wisdom **Part A 2:** 32,35,38 - Data; Table - Information; Getting hotter - Knowledge; Plan shade - Wisdom **Part B 1:** b) 16, a) Information, c) Up **Part B 2:** raw data, information, insight **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** insight, organise, why, ladder **Exercise C:** 1-F 2-T 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Data; Insight; Wisdom **Story next:** Any sensible choice; best: The team analyses library borrowing data **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
2. Machines That Predict	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Predict house price: Regression; Spam or not spam: Classification; Tomorrow's temperature: Regression; Cat or dog photo: Classification; Arrival time in minutes: Regression **Part A 2:** regression, classification, accuracy **Part B 1:** House price - Regression; Spam - Classification; Arrival - Regression; Healthy leaf - Classification **Part B 2:** 1-Collect past data, 2-Train the model, 3-Predict new cases, 4-Measure accuracy **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** perfect, error, regression, classification **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Regression; Classification; Accuracy **Story next:** Any sensible choice; best: Riya predicts canteen demand next **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

3. Python in Action	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-T 4-F 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 14, 90, 6 **Part A 2:** { } - Dictionary; def - Defines function; while - Repeats while true; return - Sends result back **Part B 1:** Circle: d = {"a": 1}, def f(x):, while n > 0: **Part B 2:** def, return, while **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** key, parameters, shows, false **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Parameter; return; Key **Story next:** Any sensible choice; best: Kabir adds a chart of house points **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
4. Solve It with Code	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Decompose - Split project; Pattern - Every Monday crowded; Abstraction - Metro map; Algorithm - Steps to make tea **Part A 2:** marks = 50: Useful; marks = 0: Useful; marks = 100: Useful; marks = pizza (no check needed): Not useful **Part B 1:** {}, +=, max **Part B 2:** Child's own work **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** abstraction, recognition, steps, cases **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Algorithm; Abstraction; Edge cases **Story next:** Any sensible choice; best: They optimise the evening route too **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
5. AI for Our Planet	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Floods - Early phone warnings; Fires - Smoke spotted from space; Pollution - Pollution forecasts; Drought - Smart irrigation **Part A 2:** climate, satellite, sustainable **Part B 1:** Cycling to school: Green; Lights on in empty rooms: Not green; Recycling paper: Green; Long showers daily: Not green; Planting trees: Green **Part B 2:** Circle: Solar, Wind, Hydro, Geothermal **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** solar, forests, language, energy **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Satellite; ISRO; Early warning **Story next:** Any sensible choice; best: Kabir builds a weather station at school **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

6. AI for Public Good	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Circle: Language translation, Reading app for the blind, Crop advice, Disaster planning **Part A 2:** public good, inclusive, accessible **Part B 1:** Blind student - Reads text aloud; Farmer - Crop advice; Different language - Translation; Cyclone - Relief planning **Part B 2:** Child's own work **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** behind, read, phones, errors **Exercise C:** 1-T 2-F 3-T 4-T **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Bhashini; Accessible; Good **Story next:** Any sensible choice; best: Riya helps add more dialects **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
7. Design Your AI Future	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Finish 10 Python lessons by March: SMART; Be good at computers: Vague; Run 2 km daily for 30 days: SMART; Become famous: Vague **Part A 2:** 1-Set a goal, 2-Learn skills, 3-Build projects, 4-Show your work **Part B 1:** S - Specific; M - Measurable; A - Achievable; T - Time-bound **Part B 2:** Child's own work **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** Measurable, actions, learning, future **Exercise C:** 1-F 2-T 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Mentor; SMART; Lifelong learning **Story next:** Any sensible choice; best: Kabir sets a SMART goal for maths **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
8. Skill-to-Career Map	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-T 6-F **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Statistics - Data scientist; Design - UX designer; Biology + code - Bioinformatics; Law + ethics - AI policy **Part A 2:** interdisciplinary, upskill, T-shaped **Part B 1:** Coding: Tech; Teamwork: People; Data analysis: Tech; Public speaking: People; Listening: People **Part B 2:** Child's own work **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** doors, field, people, India **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** T; Upskilling; Interdisciplinary **Story next:** Any sensible choice; best: The team builds a medicine-reminder app **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

9. AI Research Room	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** 1-Ask a question, 2-Search sources, 3-Verify facts, 4-Cite & write **Part A 2:** Government data site: Trustworthy; Anonymous forward: Check carefully; Published textbook: Trustworthy; AI answer without sources: Check carefully; Science journal: Trustworthy **Part B 1:** Citation - Credit to a source; Hypothesis - A testable guess; Hallucination - AI inventing facts; Verify - Check if true **Part B 2:** Child's own work **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** question, link, Scholar, support **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Hypothesis; Hallucination; Citation **Story next:** Any sensible choice; best: Riya presents at a science fair **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
10. AI Creator Studio	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Podcast - Audio; E-magazine - Text; Explainer - Video; Infographic - Image **Part A 2:** Child's own work **Part B 1:** 1-Plan the message, 2-Create media, 3-Review & improve, 4-Credit & share **Part B 2:** multimodal, infographic, original **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** video, words, data, credit **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Multimodal; Infographic; Credits **Story next:** Any sensible choice; best: They create a campaign on electricity **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.
11. Think Before You Believe	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-T 3-F 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Filter bubble - Feed shows one side; Confirmation bias - Ignoring facts you dislike; Clickbait - You won't believe this; Hallucination - AI invents a source **Part A 2:** clickbait, bubble, bias **Part B 1:** Circle: SHOCKING!!!, You won't believe..., Click to find out!, ALL CAPS anger **Part B 2:** Compares two sources: Critical; Shares because it agrees with me: Not critical; Asks "Who made this?": Critical; Trusts AI without checking: Not critical **Exercise A:** 1-a 2-b 3-c 4-a **Exercise B:** biases, like, open, decide **Exercise C:** 1-T 2-F 3-T 4-F **Exercise D:** 1-b 2-d 3-a 4-c **Exercise E:** Clickbait; Filter bubble; Confirmation **Story next:** Any sensible choice; best: They run a media literacy workshop **Exercise G (case):** 1-a 2-b 3-c **Think deeper:** Open answer; reward reasoning and examples.

12. Digital Footprints & AI

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-T 6-F
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Science fair certificate: Smart; Live location at home: Risky; Artwork you made: Smart; Angry rant about a classmate: Risky; Photo of exam hall ticket: Risky
Part A 2: footprint, profiling, reputation
Part B 1: Live location - Turn it off; Angry comment - Delete it; Public profile - Make it private; Achievement - Share it proudly
Part B 2: Child's own work
Exercise A: 1-a 2-b 3-c 4-a
Exercise B: post, ads, private, forgets
Exercise C: 1-T 2-F 3-T 4-F
Exercise D: 1-b 2-d 3-a 4-c
Exercise E: Digital footprint; Profiling; Privacy settings
Story next: Any sensible choice; best: Riya builds a positive online portfolio
Exercise G (case): 1-a 2-b 3-c
Think deeper: Open answer; reward reasoning and examples.

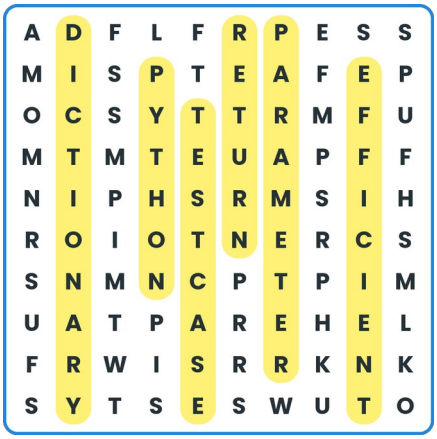
Unit reviews & practice test

AI DISCOVER: 1-a 2-b 3-c 4-a 5-b 6-c • AI CONNECT: 1-a 2-b 3-c 4-a 5-b 6-c • AI SOLVE: 1-a 2-b 3-c 4-a 5-b 6-c • AI RISE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CREATE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CARE: 1-a 2-b 3-c 4-a 5-b 6-c

Olympiad Practice Test: 1-a 2-b 3-c 4-a 5-b 6-c 7-a 8-b 9-c 10-a 11-b 12-c 13-a 14-b 15-c 16-a 17-b 18-c 19-a 20-b

WORD SEARCH ANSWERS

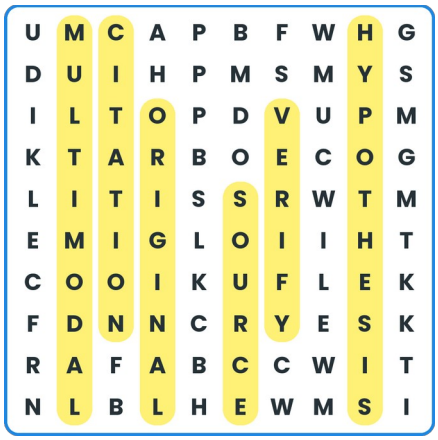
AI DISCOVER



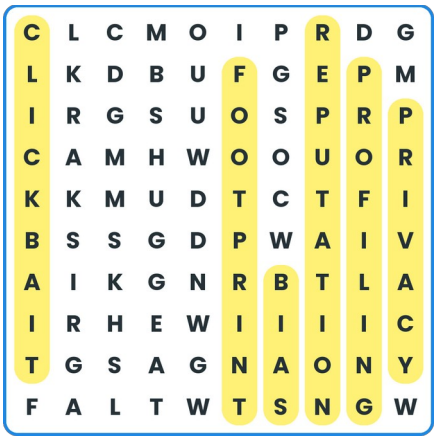
AI CONNECT

AI SOLVE

AI RISE



AI CREATE



AI CARE

MY NOTES

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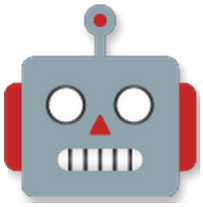

About this book

A hands-on AI skills book for senior learners of Class 9, following the 6-month Global AI Olympiad journey.

- 💬 Tech Talk: concepts as a group chat
- 🧪 Demo Lab: hands-on real AI tools
- 📚 AI Chronicles: 12 illustrated stories
- 📝 Standard, case-based & HOTS questions
- 🏆 Olympiad practice test & certificate

AI Basics | **Communicate with AI** | **AI Applications**

AI & Career | **AI Tools** | **Responsible AI**

Think | Explore | Create | A Better Tomorrow

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